Lecture 8: On "Preferences" and Preference Data

RLHFBOOK.COM

Nathan Lambert

Post-training course.
Chapters 10 & 11.

How do we hillclimb on capabilities we don't really know how to score?

Is it easier to write a good demonstration or decide between a good and a bad example?

Which poem is better? Which model?

Same prompt to two frontier models (Claude 3.7 Sonnet and ChatGPT with GPT-4o, Feb 2025): “Write me a short poem about an optimistic goldfish.” Which is better – and which came from which?

ASSISTANT (A)

The Optimistic Goldfish

A little goldfish, bright and bold,
Swam in circles, dreams untold.
Though the bowl was small and round,
Endless wonders still abound.

“The world is vast,” she’d always say,
“I’ll find new joys in every day!”
No walls could dim her hopeful view,
For in her heart, the ocean grew.

ASSISTANT (B)

The Optimistic Goldfish

In a bowl of curved glass,
Swims a goldfish, bright and bold.
Three seconds of memory, they say,
Yet each circuit feels brand new and gold.

“Today might bring fresh flakes,
Or perhaps a castle tall.
What adventure awaits beyond this wall?”
The optimist in golden hue.

Whenpreferencereplacescorrectness

Most of post-training so far (SFT, RLVR) has had a notion of correct – a math answer, a passing test, a verifiable reward. Preferences are what we reach for when there is **no right answer** .

E.g. Which poem is better? Which explanation is clearer? Which tone is kinder?

Preferences came to be the core of RLHF, as the alignment community looked to optimize for complicated objectives like “human values.”

This lecture

We trace where “preferences” came from, argue why it’s an **imperfect problem formulation**, then dig into **preference data** (chapter 11) used for today’s models.

The plan

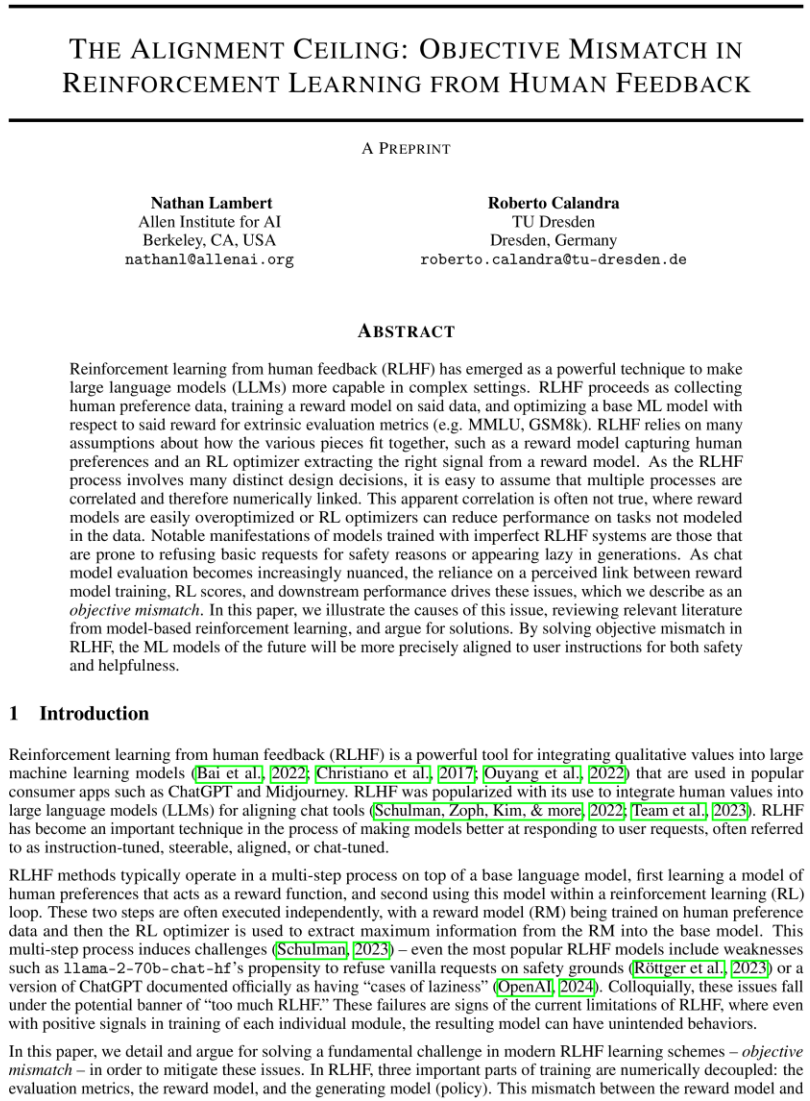
1. A short **history** of preferences – and why it’s an **imperfect** formulation (chapter 10)
2. **Preference data** – the trade-offs in practice (chapter 11)
3. **Open questions** (some shared with synthetic data)

Aside: The "objective mismatch" lens

A recurring way to think about post-training: we optimize a **proxy** objective that isn't the one we actually care about.

- **Model-based RL:** the dynamics model is trained for prediction accuracy, not control performance – so a "better" model can yield a worse policy (Lambert et al., 2020).
- **RLHF:** the reward model is trained for preference-classification accuracy, not downstream policy quality – the same mismatch, one level up (Lambert & Calandra, 2023).

Reward-model accuracy (RewardBench-style) is a proxy for a proxy! Keep asking what you are really optimizing.



THE ALIGNMENT CEILING: OBJECTIVE MISMATCH IN REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

A PREPRINT

Nathan Lambert
Allen Institute for AI
Berkeley, CA, USA
nathanl@allenai.org

Roberto Calandra
TU Dresden
Dresden, Germany
roberto.calandra@tu-dresden.de

arXiv:2311.00168v2 [cs.LG] 2 Feb 2024

ABSTRACT

Reinforcement learning from human feedback (RLHF) has emerged as a powerful technique to make large language models (LLMs) more capable in complex settings. RLHF proceeds as collecting human preference data, training a reward model on said data, and optimizing a base ML model with respect to said reward for extrinsic evaluation metrics (e.g. MMLU, GSM8k). RLHF relies on many assumptions about how the various pieces fit together, such as a reward model capturing human preferences and an RL optimizer extracting the right signal from a reward model. As the RLHF process involves many distinct design decisions, it is easy to assume that multiple processes are correlated and therefore numerically linked. This apparent correlation is often not true, where reward models are easily overoptimized or RL optimizers can reduce performance on tasks not modeled in the data. Notable manifestations of models trained with imperfect RLHF systems are those that are prone to refusing basic requests for safety reasons or appearing lazy in generations. As chat model evaluation becomes increasingly nuanced, the reliance on a perceived link between reward model training, RL scores, and downstream performance drives these issues, which we describe as an *objective mismatch*. In this paper, we illustrate the causes of this issue, reviewing relevant literature from model-based reinforcement learning, and argue for solutions. By solving objective mismatch in RLHF, the ML models of the future will be more precisely aligned to user instructions for both safety and helpfulness.

1 Introduction

Reinforcement learning from human feedback (RLHF) is a powerful tool for integrating qualitative values into large machine learning models (Bai et al., 2022; Christiano et al., 2017; Ouyang et al., 2022) that are used in popular consumer apps such as ChatGPT and Midjourney. RLHF was popularized with its use to integrate human values into large language models (LLMs) for aligning chat tools (Schulman, Zoph, Kim, & more, 2022; Team et al., 2023). RLHF has become an important technique in the process of making models better at responding to user requests, often referred to as instruction-tuned, steerable, aligned, or chat-tuned.

RLHF methods typically operate in a multi-step process on top of a base language model, first learning a model of human preferences that acts as a reward function, and second using this model within a reinforcement learning (RL) loop. These two steps are often executed independently, with a reward model (RM) being trained on human preference data and then the RL optimizer is used to extract maximum information from the RM into the base model. This multi-step process induces challenges (Schulman, 2023) – even the most popular RLHF models include weaknesses such as `llama-2-70b-chat-hf`'s propensity to refuse vanilla requests on safety grounds (Röttger et al., 2023) or a version of ChatGPT documented officially as having "cases of laziness" (OpenAI, 2024). Colloquially, these issues fall under the potential banner of "too much RLHF." These failures are signs of the current limitations of RLHF, where even with positive signals in training of each individual module, the resulting model can have unintended behaviors.

In this paper, we detail and argue for solving a fundamental challenge in modern RLHF learning schemes – *objective mismatch* – in order to mitigate these issues. In RLHF, three important parts of training are numerically decoupled: the evaluation metrics, the reward model, and the generating model (policy). This mismatch between the reward model and

Dynamics Model

마르코프 의사결정 과정(MDP):

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma), \qquad s_{t+1} \sim P(\cdot \mid s_t, a_t)$$

환경의 전이 확률을 근사하는 Dynamics Model:

$$\hat{P}_\theta(s_{t+1} \mid s_t, a_t) \approx P(s_{t+1} \mid s_t, a_t)$$

결정론적 표현의 특수한 경우:

$$\hat{s}_{t+1} = f_\theta(s_t, a_t)$$

Dynamics Model vs. Policy Control

$$\max_{\theta} \sum_{i=1}^{N} \log p_\theta(s'_i \mid s_i, a_i)$$

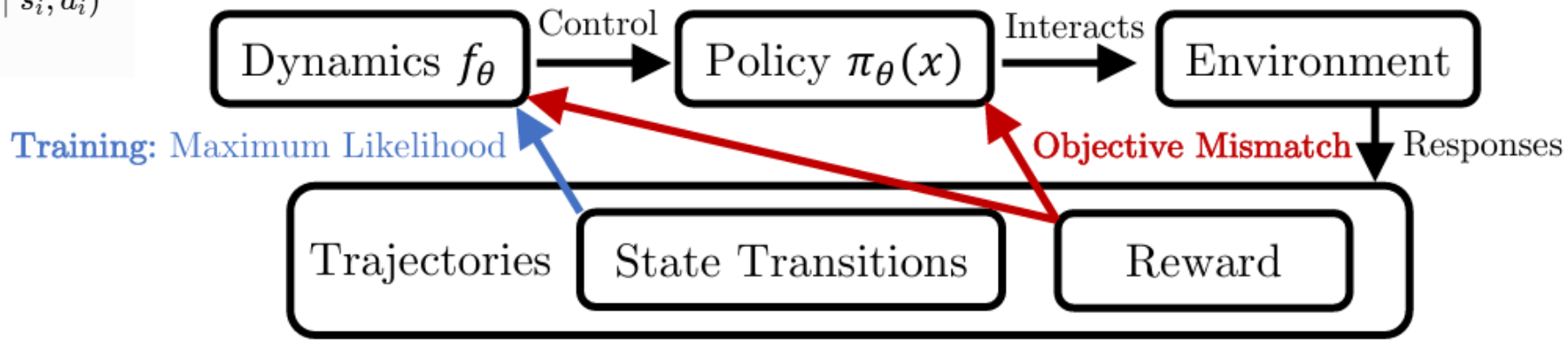
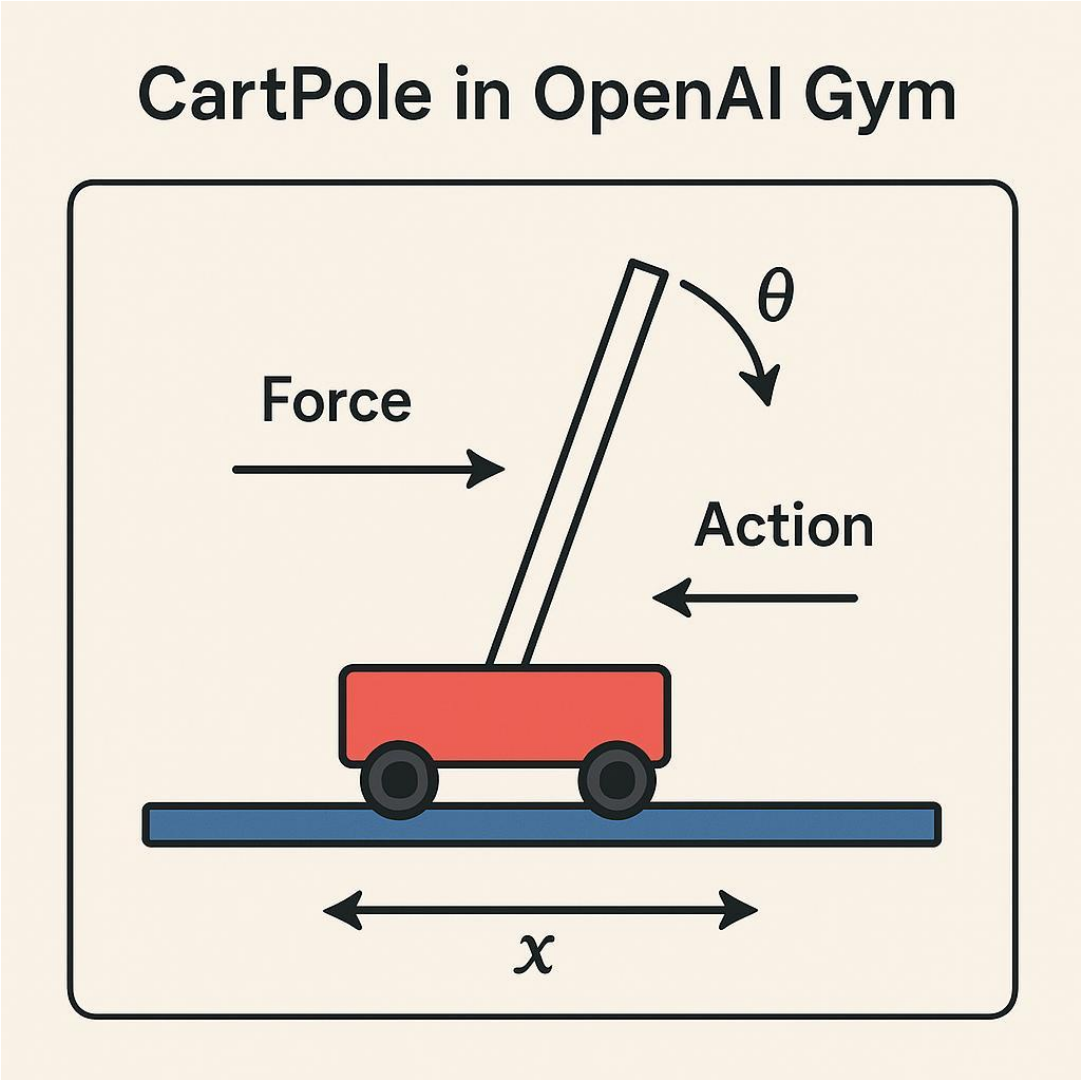


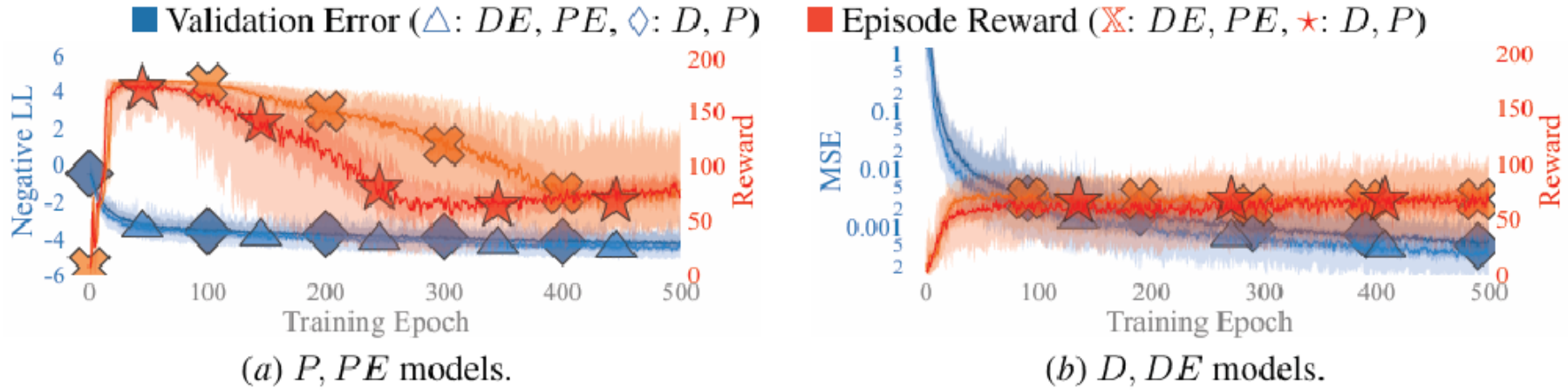
Figure 1: Objective mismatch in MBRL arises when a model is trained to maximize the likelihood but then used for control to maximize a reward signal not considered during training.

Training: $\arg\max_{\theta} \sum_{i=1}^{N} \log p_\theta(s'_i|s_i, a_i)$, **Control:** $\arg\max_{a_{t:t+T}} \mathbb{E}_{\pi_\theta(s_t)} \sum_{i=t}^{t+T} r(s_i, a_i)$

Dynamics Model



Dynamics Model: Objective Mismatch



(a) P, PE models. (b) D, DE models.

Figure 4: The reward when re-evaluating the controller at each dynamics model training epoch for different dynamics models, $M = 50$ per model type. Even for the simple cartpole environment, D, DE fail to achieve full performance, while P, PE reach higher performance but eventually over-fit to available data. The validation loss is still improving slowly at 500 epochs, not yet over-fitting.

RHLF: Objective Mismatch

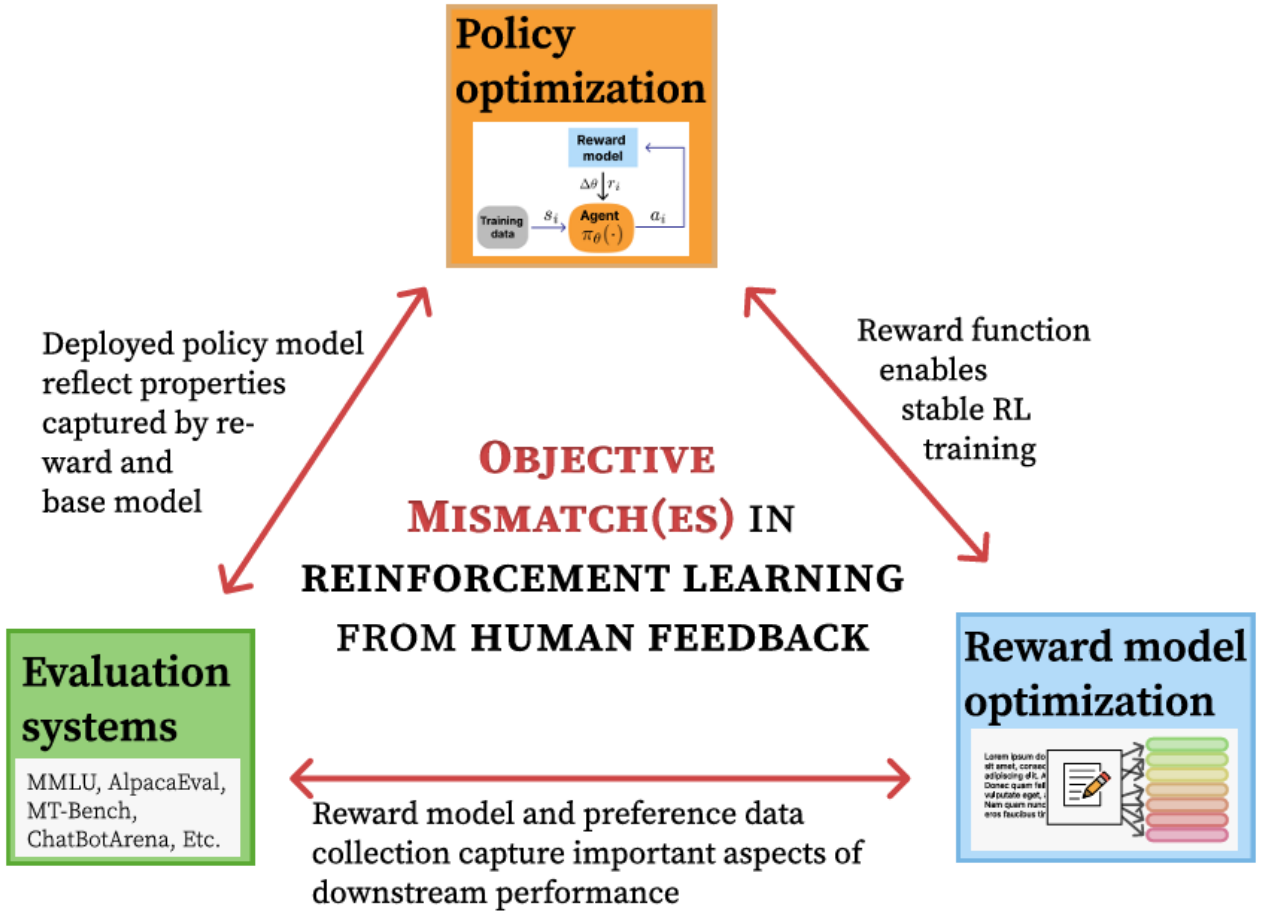


Figure 1: The three links causing objective mismatch in RLHF: Reward model training ↔ policy model training, reward model training ↔ evaluation tools, and policy model training ↔ evaluation tools, as discussed in Sec. 4.1.

RHLF: Objective Mismatch

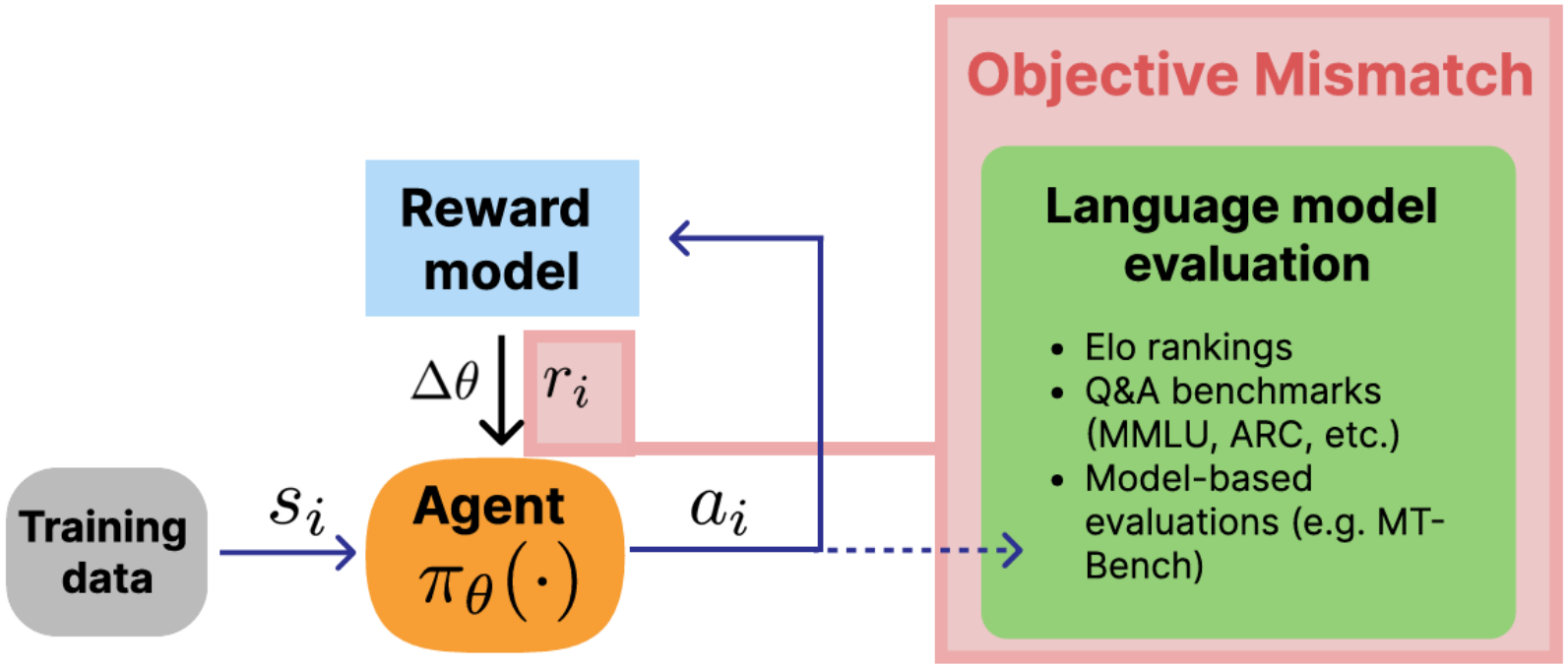
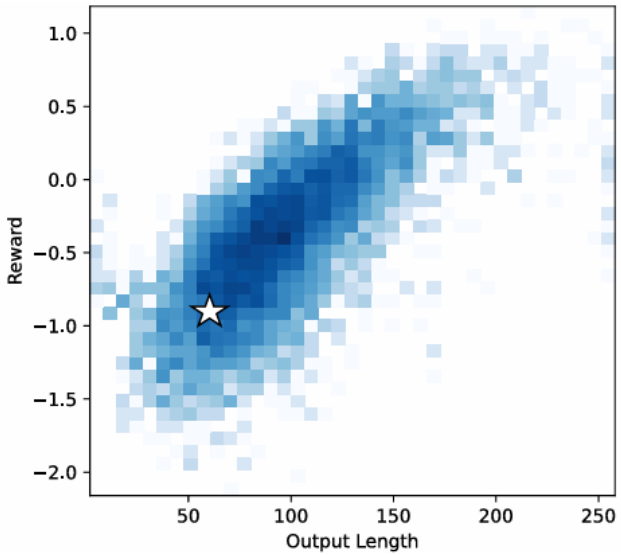


Figure 2: An illustration of where the objective mismatch issue emerges within the RL optimization phase of RLHF. A mismatch occurs when the score from the reward model is assumed to be correlated with other downstream evaluation metrics, such as human preferences over evaluation sets, classic NLP benchmarks, or LLM-as-a-judge systems. Compared to traditional RL problems, RLHF does not have the canonical form of an *environment*, which indirectly maps to the training data with a reward model, but does not capture the same properties.

RHLF: Longer is Better



Question: *Why don't adults roll off the bed?*

☆ **SFT (Before); 59 tokens**

Adults typically do not roll off of the bed because they have developed the muscle memory to keep their bodies from involuntarily moving during sleep and maintaining proper posture.

RLHF (After); 243 tokens: **Similar output, but much longer / more details**

Adults generally do not roll off of the bed because they have developed muscle control and awareness which helps them to remain upright while sleeping. Additionally, most adults find it uncomfortable or even painful to move around in their sleep, so rolling off the bed would be difficult without waking up first. There may also be safety concerns such as falling out of bed and potentially hurting oneself if one were to lose balance or fall from the ...

Figure 1: Log-scaled heatmap of lengths of SFT outputs vs. learned reward model scores for WebGPT (left). The graph shows that reward scores are strongly correlated with length. RLHF with these consistently leads to longer outputs (right).

A Long Way to Go: Investigating Length Correlations in RLHF, COLM 2024

RHLF: Objective Mismatch

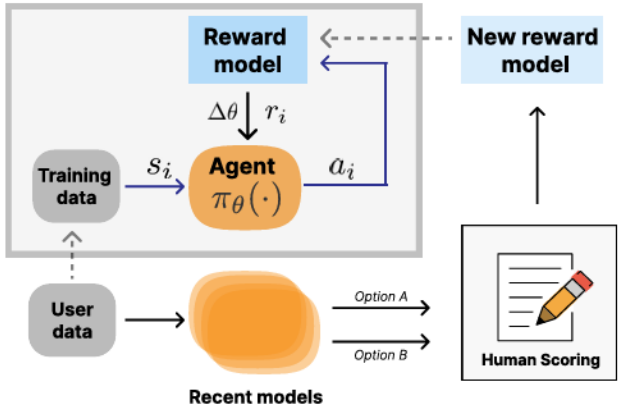


Figure 5: The outer loop of RLHF is the process to train the next reward model for RL to improve performance is areas of user interest. This setup induces additional complexity in objective mismatch in RLHF.

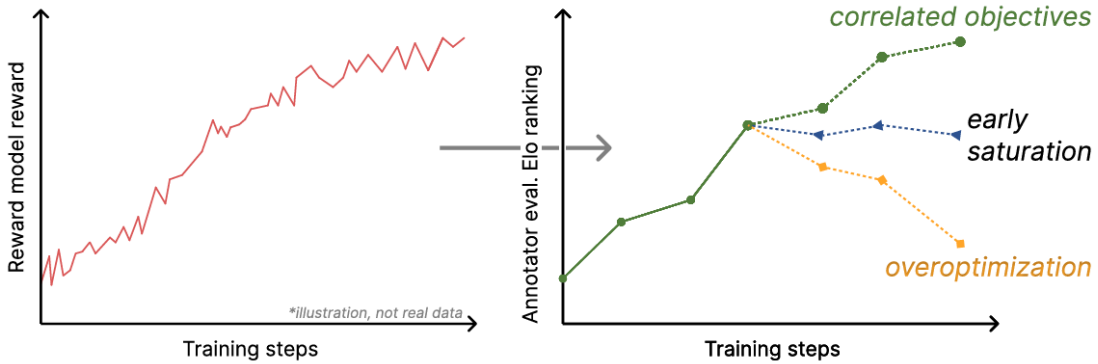
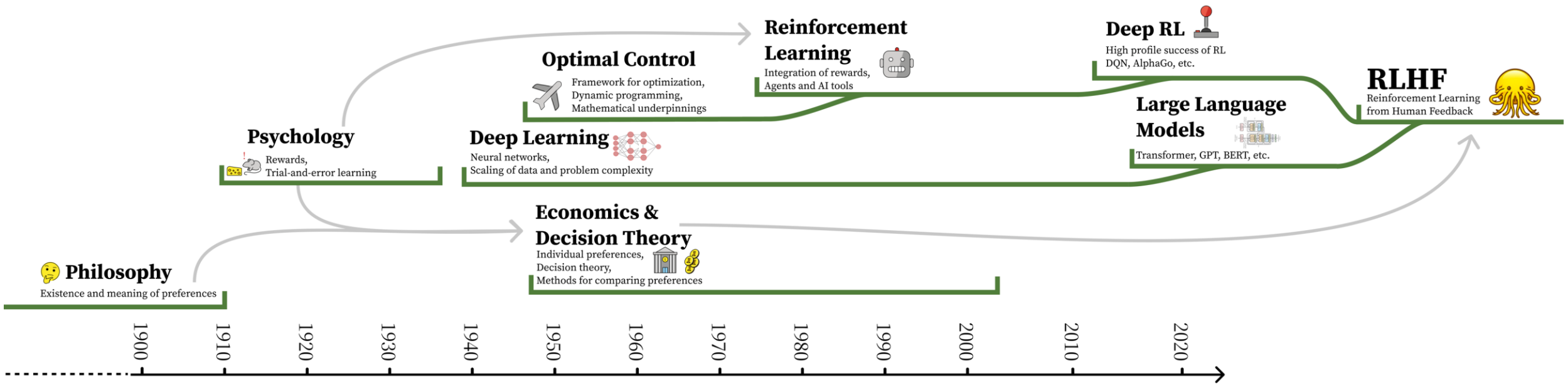


Figure 4: Illustrating the most likely visualization of objective mismatch in RLHF, the link between policy training and downstream evaluation. Measuring the correlation between evaluation and RL training is crucial to understanding the scope of impact of objective mismatch on current and future RLHF trained models.

Part 1: A short history of optimizing or measuring preferences

Many fields converged into "RLHF"



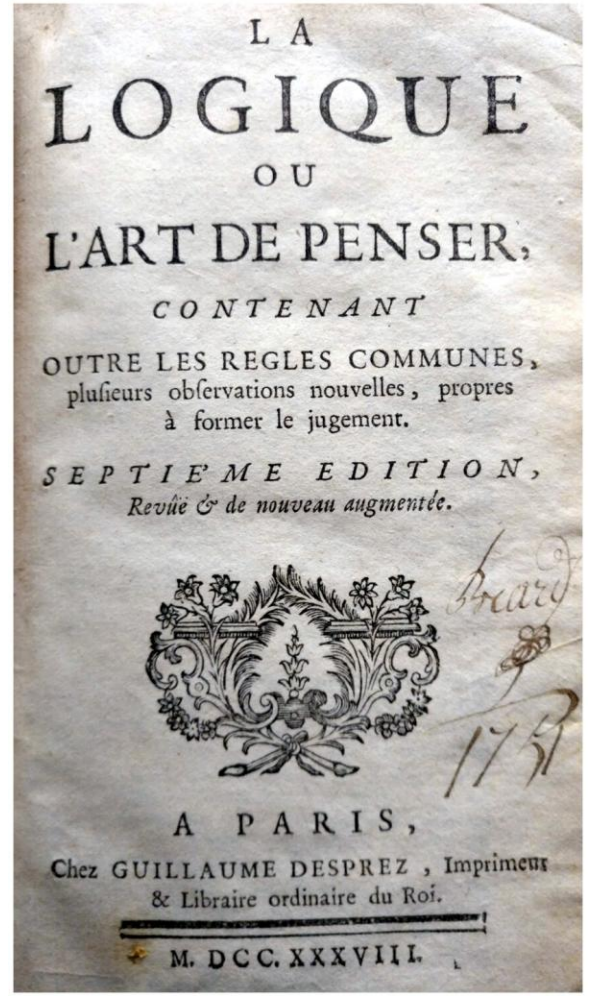
Lambert et al., 2023

Utility, from logic to a number

The idea that choices can be scored is old. The common thread: the idea that human wants can be reduced to a single measure:

- **Port Royal Logic** (1662): decision quality = outcome weighted by its probability (Arnauld, 1662)
- **Bentham's hedonic calculus** (early 1800s): weigh all of life on one complicated, but common scale (Bentham, 1823)
- **Ramsey**, Truth and Probability (1931): first to quantify preference and belief together as the way that individuals make probabilistic decisions (Ramsey, 2016)

> "To judge what one must do to obtain a good or avoid an evil, it is necessary to consider not only the good and evil in itself, but also the probability that it


LA
LOGIQUE
OU
L'ART DE PENSER,
CONTENANT
OUTRE LES REGLES COMMUNES,
plusieurs observations nouvelles, propres
à former le jugement.
SEPTIE'ME EDITION,
Revûe & de nouveau augmentée.
A PARIS,
Chez GUILLAUME DESPREZ, Imprimeur
& Libraire ordinaire du Roi.
M. DCC. XXXVIII.

La Logique, ou l'Art de Penser (the Port Royal Logic), 1662

Von Neumann-Morgenstern utility theorem (1947)

Von Neumann & Morgenstern (1947): if your preferences obey a few axioms (completeness, transitivity, continuity, independence), they can be represented by a single **utility function**, and rational choice = maximizing **expected utility** (Von Neumann & Morgenstern, 1947).

This is the result RLHF leans on to justify fitting a scalar reward.

In RLHF, essentially none of those ifs hold:

- preferences **drift** during and after labeling
- they're **context- and framing-dependent**
- at high complexity they can be **intransitive**
- and they're **multidimensional**, squashed into one number



John von Neumann (Los Alamos badge photo, public domain)

One scalar preference, many features

A reward model compresses, into a single number, all of:

- helpfulness, honesty, harmlessness, tone, format, length, taste…
- the annotator's psychology, culture, and the interface they used
- whatever the framing of the comparison nudged them toward

We then optimize hard against that number, which is impossible to ever do perfectly! There will always be trade-offs.

Artificial Intelligence: A Modern Approach

Outline

- Combining Beliefs and Desires under Uncertainty Utilities
- The Basis of Utility Theory
- Utility Functions
- Multiattribute Utility Functions
- Decision networks
- Value of information
- Unknown Preferences

핵심 질문
무엇이 일어날 것이며, 무엇을 원하는가?
합리적인 선택의 조건은 무엇인가?
결과의 가치를 어떻게 숫자로 표현하는가?
여러 기준을 어떻게 동시에 고려하는가?
확률·행동·효용을 어떻게 결합하는가?
새로운 정보를 얻는 것이 얼마나 가치 있는가?
선호 자체를 모를 때는 어떻게 하는가?

왜 LLM은 글을 잘 쓰지 못할까?

Artificial Intelligence: A Modern Approach, Copyright © 2021 Pearson Education, Inc. All Rights Reserved

Artificial Intelligence: A Modern Approach

An agent assigns a probability $P(s)$ to each possible current state s. There may also be uncertainty about the action outcomes; the transition model is given by $P(s' \mid s, a)$,

$$P(\text{RESULT}(a) = s') = \sum_s P(s)P(s' \mid s, a) .$$

The **expected utility** of an action given the evidence, $EU(a)$, is just the average utility value of the outcomes, weighted by the probability that the outcome occurs:

$$EU(a) = \sum_{s'} P(\text{RESULT}(a) = s')U(s') .$$

The principle of **maximum expected utility (MEU)** says that a rational agent should choose the action that maximizes the agent's expected utility:

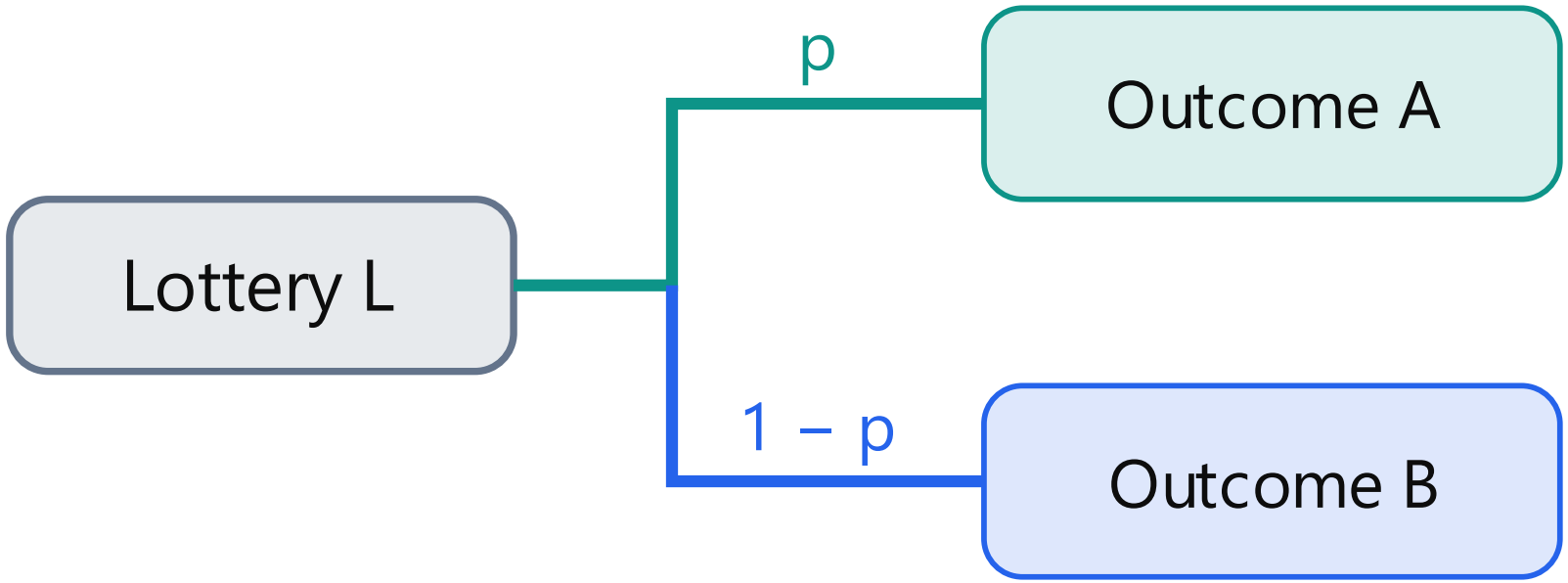
$$action = \underset{a}{\text{argmax}}\, EU(a) .$$

If agent acts so as to maximize a utility function that correctly reflects the performance measure, then the agent will achieve the highest possible performance score (averaged over all the possible environments).

Artificial Intelligence: A Modern Approach, Copyright © 2021 Pearson Education, Inc. All Rights Reserved

The Basis of Utility Theory

기호	의미
$A \succ B$	A를 B보다 선호
$A \sim B$	A와 B를 동등하게 선호
$A \succeq B$	A를 B보다 덜 선호하지 않음



Artificial Intelligence: A Modern Approach, Copyright © 2021 Pearson Education, Inc. All Rights Reserved

Artificial Intelligence: A Modern Approach

Rational preferences

Idea: preferences of a rational agent must obey constraints.
Rational preferences ⇒
behavior describable as maximization of expected utility

Constraints:

Orderability

$(A <; B) \lor (B <; A) \lor (A \sim B)$

Transitivity

$(A <; B) \land (B <; C) \Rightarrow (A <; C)$

Continuity

$A <; B <; C \Rightarrow \exists\, p \quad [p, A;\ 1 - p, C] \sim B$

Substitutability

$A \sim B \Rightarrow [p, A;\ 1 - p, C] \sim [p, B;\ 1 - p, C]$

Monotonicity

$A <; B \Rightarrow (p \geq q \Leftrightarrow [p, A;\ 1 - p, B] \underset{\sim}{<;} [q, A;\ 1 - q, B])$

공리	쉬운 의미	예시
완전성 (Completeness)	두 선택지를 비교할 수 있다.	A가 좋거나, B가 좋거나, 둘이 비슷하다.
추이성(Transitivity)	선호의 순서가 일관적이다.	A보다 B, B보다 C가 좋다면 C보다 A가 좋다고 할 수는 없다.
연속성(Continuity)	중간 수준의 선택은 최고·최악 결과를 섞은 확률적 선택과 동등해질 수 있다.	확실한 보통 수준의 결과와 최고·최악 결과가 나오는 복권 사이에 무차별한 확률이 존재한다.
독립성 (Independence)	동일한 제3의 선택지를 같은 비율로 섞어도 기존 선호가 유지된다.	A를 B보다 선호한다면, A와 C의 혼합도 같은 비율의 B와 C 혼합보다 선호한다.

이 조건들이 확률적 선택지(복권)에 대한 선호에서 성립하면, 다음과 같이 기대 효용을 표현하는 함수가 존재합니다.

$$U(L) = \sum_i p_i u(o_i)$$

- o_i: 가능한 결과
- p_i: 해당 결과가 발생할 확률
- $u(o_i)$: 해당 결과의 효용

Artificial Intelligence: A Modern Approach, Copyright © 2021 Pearson Education, Inc. All Rights Reserved

Maximizing expected utility

Theorem (Ramsey, 1931; von Neumann and Morgenstern, 1944): Given preferences satisfying the constraints there exists a real-valued function U such that

$$U(A) \geq U(B) \quad \Leftrightarrow \quad A \succsim B$$
$$U([p_1, S_1; \ldots; p_n, S_n]) = \sum_i p_i U(S_i)$$

MEU principle:

Choose the action that maximizes expected utility

Note: an agent can be entirely rational (consistent with MEU) without ever representing or manipulating utilities and probabilities

E.g., a lookup table for perfect tictactoe

$$\boxed{U(L) = \sum_{i=1}^{n} p_i U(S_i)}$$

- L: Lottery
- p_i: 결과 S_i가 발생할 확률
- $U(S_i)$: 해당 결과의 효용

예를 들어 다음과 같은 Lottery가 있다고 가정하겠습니다.

결과	확률	효용	확률 × 효용
A	50%	100	50
B	30%	50	15
C	20%	0	0

따라서 전체 기대효용은 다음과 같습니다.

$$U(L) = 0.5(100) + 0.3(50) + 0.2(0) = 65$$

이 Lottery의 기대효용은 65입니다.

여러 선택지가 주어졌다면 기대효용이 가장 높은 선택지를 고르는 것이 MEU(Maximum Expected Utility) 원칙입니다.

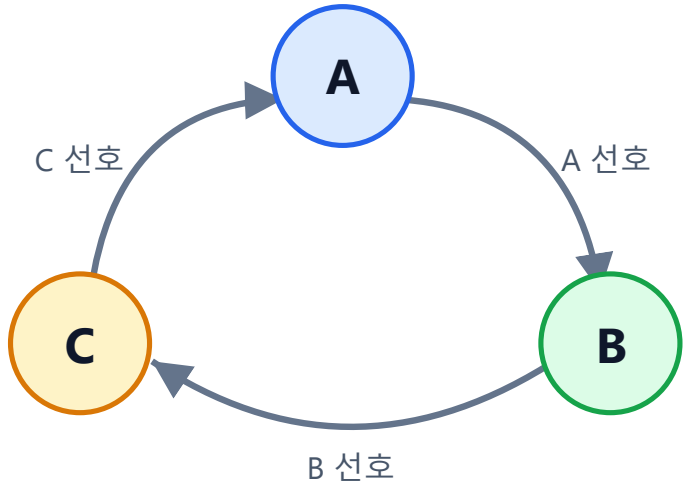
Artificial Intelligence: A Modern Approach, Copyright © 2021 Pearson Education, Inc. All Rights Reserved

Bradley-Terry (1952): Comparisons to scores

The statistical model that turns comparisons into scores – and became the backbone of reward modeling (Bradley & Terry, 1952) (see Chapter 5 / Lecture 2):

$$P(y_w \succ y_l | \ x) = \ \sigma(r(x, y_w) - \ r(x, y_l)) = \frac{e^{r(x,y_w)}}{e^{r(x,y_w)} + \ e^{r(x,y_l)}}$$

Give it pairwise human comparisons; out comes a scalar reward. This is why RLHF needs **preference data** (and where the imperfections enter).



Whereutilitytheorybreaksdown

Almost as soon as utility was formalized, social choice (the field studying how preferences should be aggregated) and economics found its limits:

- **Arrow's impossibility theorem** (1950): no voting rule aggregates individual preferences into a collective one while satisfying a few basic fairness criteria (Arrow, 1950)
- **Sen**, Behaviour, Choice and Values (1973): choice ≠ preference; revealed-preference theory is too thin (Sen, 1973)
- **Hirschman**, Against Parsimony (1984): people have preferences over their preferences – so preferences may be unmeasurable (Hirschman, 1984)



Kenneth Arrow (CC BY 3.0, via Wikimedia Commons)

Aside: Social choice for AI alignment

If aggregating preferences is the hard part, **social choice theory** is the field that studies it – and a natural lens for alignment.

Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback.

- A 2024 position paper I was part of argues social choice should guide how we aggregate diverse human feedback (Conitzer et al., 2024).
- Whose preferences count? How do you combine disagreeing labelers? ...
- (One of the rabbit holes I used to spend more time in.)

arXiv:2404.10271v2 [cs.LG] 4 Jun 2024

Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback

Vincent Conitzer [1 2] Rachel Freedman [3] Jobst Heitzig [4] Wesley H. Holliday [5] Bob M. Jacobs [6] Nathan Lambert [7] Milan Mossé [5] Eric Pacuit [8] Stuart Russell [3] Hailey Schoelkopf [9] Emanuel Tewolde [1] William S. Zwicker [10 11]

Abstract

Foundation models such as GPT-4 are fine-tuned to avoid unsafe or otherwise problematic behavior, such as helping to commit crimes or producing racist text. One approach to fine-tuning, called *reinforcement learning from human feedback*, learns from humans' expressed preferences over multiple outputs. Another approach is *constitutional AI*, in which the input from humans is a list of high-level principles. But how do we deal with potentially diverging input from humans? How can we aggregate the input into consistent data about "collective" preferences or otherwise use it to make collective choices about model behavior? In this paper, we argue that the field of *social choice* is well positioned to address these questions, and we discuss ways forward for this agenda, drawing on discussions in a recent workshop on Social Choice for AI Ethics and Safety held in Berkeley, CA, USA in December 2023.

[1]Foundations of Cooperative AI Lab, Computer Science Department, Carnegie Mellon University, Pittsburgh, USA [2]Institute for Ethics in AI, University of Oxford, Oxford, United Kingdom [3]Center for Human-Compatible AI, Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, USA [4]Potsdam Institute for Climate Impact Research, Potsdam, Brandenburg, Germany [5]Department of Philosophy, University of California, Berkeley, USA [6]Department of Philosophy and Moral Sciences, Ghent University, Ghent, Begium [7]Allen Institute for AI, Berkeley, California, USA [8]Department of Philosophy, University of Maryland, College Park, USA [9]EleutherAI [10]Department of Mathematics, Union College, Schenectady, USA [11]Murat Sertel Center for Advanced Economic Studies, Istanbul Bilgi University, Istanbul, Turkey. Correspondence to: Vincent Conitzer <conitzer@cs.cmu.edu>.

Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024. Copyright 2024 by the author(s).

1. Introduction

Recently, *reinforcement learning from human feedback* (RLHF) has become the primary strategy that leading AI companies such as OpenAI (OpenAI, 2023), Anthropic (Anthropic, 2023), Meta (Meta, 2023), and Google (Google, 2023) use to make pretrained large language models (LLMs) more capable and controllable (Christiano et al., 2017; Ziegler et al., 2019) and to align them with human values. However, RLHF faces many limitations and challenges (Casper et al., 2023; Lambert & Calandra, 2023), including unrepresentative data (Prabhakaran et al., 2021; Feffer et al., 2023), unrealistic models of human decision-making (Hong et al., 2022; Freedman et al., 2021; Siththaranjan et al., 2023; Lambert et al., 2023), and insufficient modeling of human diversity (Kirk et al., 2023; Freedman et al., 2023) which may lead to political bias (Motoki et al., 2023; Rozado, 2024). We propose that ideas from social choice theory (Arrow, 2012; Fishburn, 1973; Kelly, 1988; Brandt et al., 2015)—e.g., concerning whose preferences should be integrated into decisions and how this should be done—are needed to solve many of these open problems.

While models that are solely pretrained on internet data may produce repetitive or harmful text, RLHF trains models to follow instructions (Ouyang et al., 2022) and produce helpful and "harmless" outputs (Bai et al., 2022a) based on human judgments. RLHF gathers example outputs from a pretrained LLM or examples written by humans. Next, humans are asked to select the outputs that best meet specified criteria (such as being "helpful" or "unbiased"). These judgments, often called *preferences*, are then used to fine-tune the LLM. From a social choice perspective, this method raises several critical questions: Which humans are asked to judge outputs? What criteria do they use? How are their judgments combined? And how do their expressed judgments relate to their actual preferences?

Constitutional AI (CAI), which involves reinforcement learning from AI feedback (RLAIF), is an alternative addressing some of these questions (Bai et al., 2022b). Humans produce a "constitution" that specifies principles that the LLM is trained to align with. However, one must still

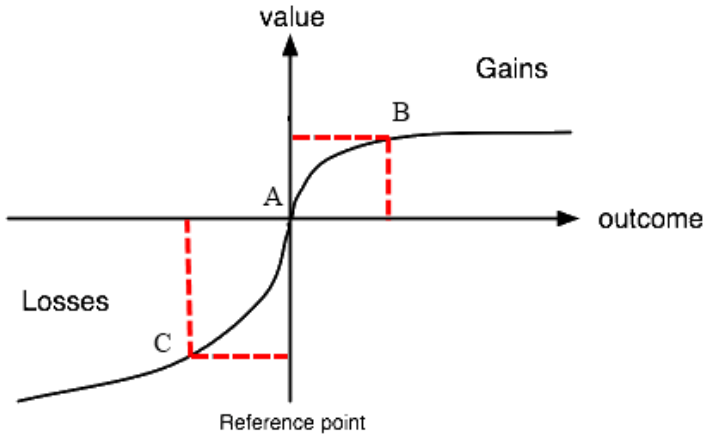
Conitzer et al., 2024

Preferences are not stable objects over time and context

From psychology and behavioral economics:

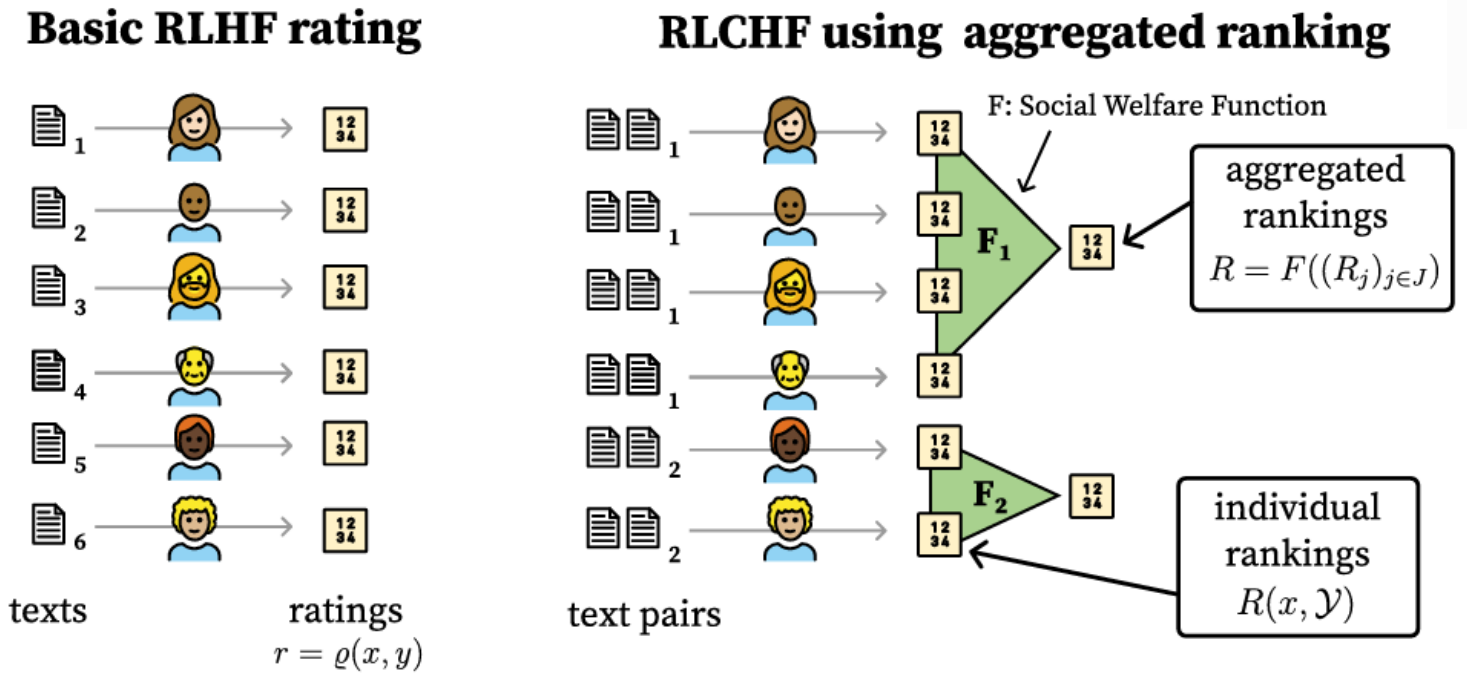
- Preferences **drift** – they change with time, mood, and experience (Pettigrew, 2019)
- Choices are shaped by situation and framing, not just an inner ranking (Gilbert et al., 2022)

A problem for "collect a label, train on it later repeatedly."



The prospect-theory value function: the same outcome is valued differently depending on the reference point – value depends on framing, not a fixed ranking (CC BY-SA 3.0, via Wikimedia Commons).

Social Choice Should Guide AI Alignment



- R_j: 평가자 j의 개별 순위
- J: 해당 답변을 평가하는 평가자 집단
- F: 여러 개인의 순위를 합치는 규칙
- R: 최종적으로 산출된 집단 순위

Figure 2. **RLCHF using aggregated rankings**. The core addition to the standard RLHF process is the call-out of an explicit social welfare function, F, which determines how preferences are aggregated.

Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback (ICML 2024)

Social Choice Should Guide AI Alignment

여기서 f_i는 평가자 i에 관한 정보입니다.

예를 들어 가상의 평가자 특성으로 다음과 같은 정보를 생각해 볼 수 있습니다.

- 간결한 설명을 선호하는지
- 전문적인 설명을 선호하는지
- 안전성을 중요하게 생각하는지

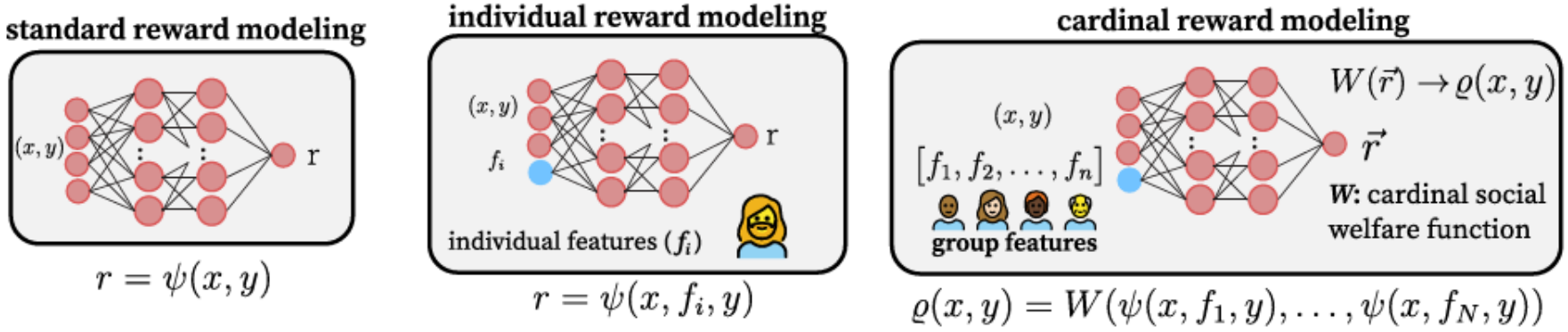


Figure 3. **RLCHF using evaluator features and aggregated ranks**. We show how an individuals' features can be used as an additional input to reward models within the RLHF process.

- $\psi(x, f_i, y)$: 평가자 i가 해당 답변에 부여할 것으로 예측한 점수
- W: 개인별 점수를 통합하는 사회후생함수
- $\varrho(x, y)$: 집단을 대표하는 최종 보상 점수

가령 두 평가자에 대한 예측이 다음과 같다고 가정하겠습니다.

평가자	답변 A	답변 B
간결함을 중시하는 평가자	9	4
정확성을 중시하는 평가자	3	9

단순 평균을 사용한다면 다음과 같습니다.

$$\varrho(A) = 6, \qquad \varrho(B) = 6.5$$

최종적으로 B의 점수가 더 높게 나옵니다.

하지만 다른 집계 규칙이나 가중치를 사용한다면 결과가 달라질 수 있습니다.

Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback (ICML 2024)

Social Choice Should Guide AI Alignment

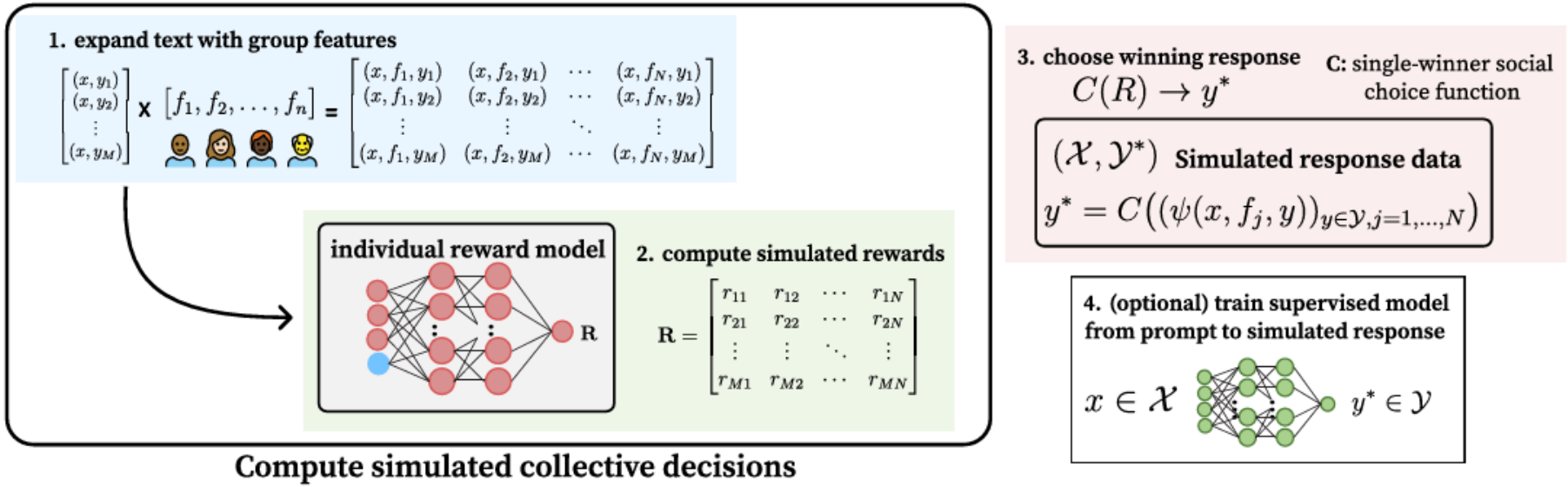


Figure 4. **Supervised Learning from Simulated Collective Decisions**. We show that with an individual or cardinal reward model, as presented in Figure 3, responses y to a prompt x can be simulated. This process expands the scope of studying preferences within RLHF and opens future work on personalization and other topics.

4. Step 4 — Optional Supervised Learning

최종 선택된 답변 y^*를 이용하면 새로운 지도학습 데이터를 만들 수 있습니다.

$$\mathcal{D}_{\text{SFT}} = \{(x, y^*)\}$$

이를 활용해 SFT(Supervised Fine-Tuning)를 진행하는 것입니다.

예를 들어 질문이 "강화학습을 설명해줘"이고, 집단적 선택 결과가 답변 B였다면 다음과 같은 학습 데이터를 만듭니다.

- Input: 강화학습을 설명해줘.
- Target: 집단이 선택한 답변 B

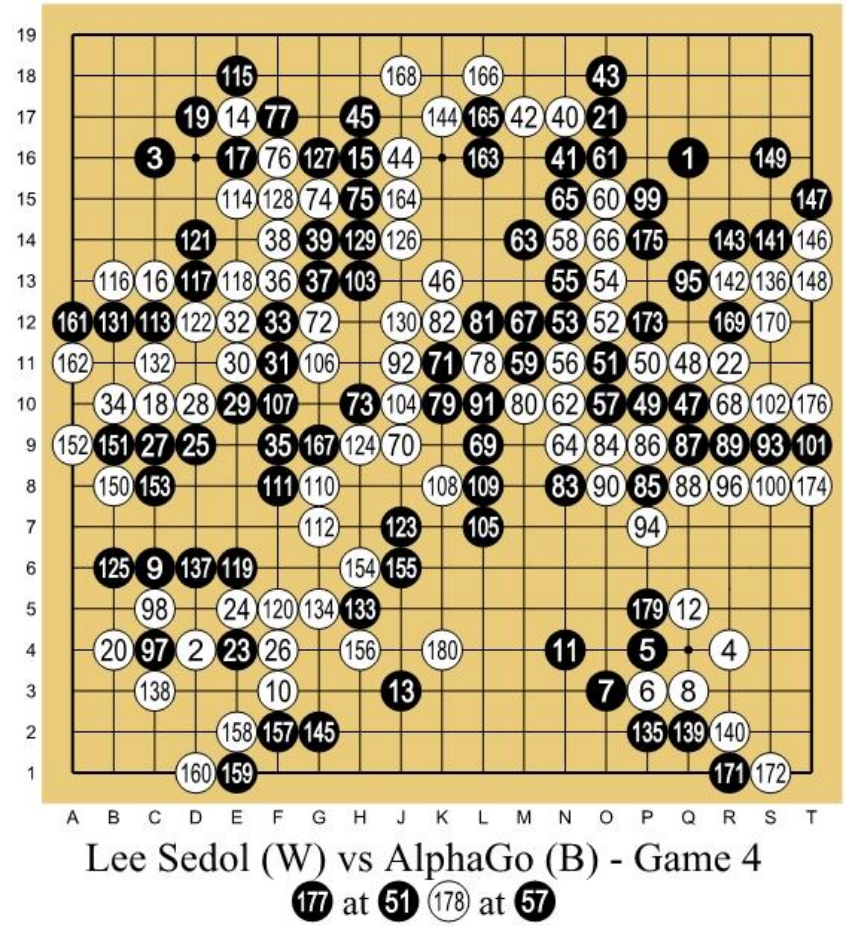
Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback (ICML 2024)

The other root: Optimal control & RL

In parallel, a machinery for optimizing a reward matured:

- **Bellman** (1957): MDPs and dynamic programming (Bellman, 1957) $\Rightarrow$ origins of "optimal" control / control theory
- **Sutton** (1988): temporal-difference learning for credit assignment (Sutton, 1988) (instead of learning from a lot of past data)
- **Watkins** (1992): Q-learning (Watkins & Dayan, 1992) (learning value)
- **DQN** (2013): deep RL at scale (Mnih et al., 2013); **AlphaGo / AlphaZero** (2017): mastery from self-play (Silver et al., 2017). "RL works"

Note: these guarantees assume a **single, closed-form reward** !



Lee Sedol vs AlphaGo, Game 4 (2016) (CC BY-SA 4.0, via Wikimedia Commons)

Costs ≠ rewards ≠ preferences

These three are **ontologically different**, and the emergence of modern post-training treated them as interchangeable (Lambert et al., 2023).

- **Costs** come from control: physical, measurable, given, and often have clear optimality/bounds.
- **Rewards** are from psychology and an RL convenience: a scalar signal to maximize.
- **Preferences** are human, relational, and unstable – not obviously a scalar at all.

Reducing preferences to rewards made the optimization format tractable, but is the root cause of many of the unsolvable "biases" in RLHF and preference data.

Inverse RL? RHLF?

DEEP LEARNING QUANTIFIES PATHOLOGISTS' VISUAL PATTERNS FOR WHOLE SLIDE IMAGE DIAGNOSIS (PEAN)

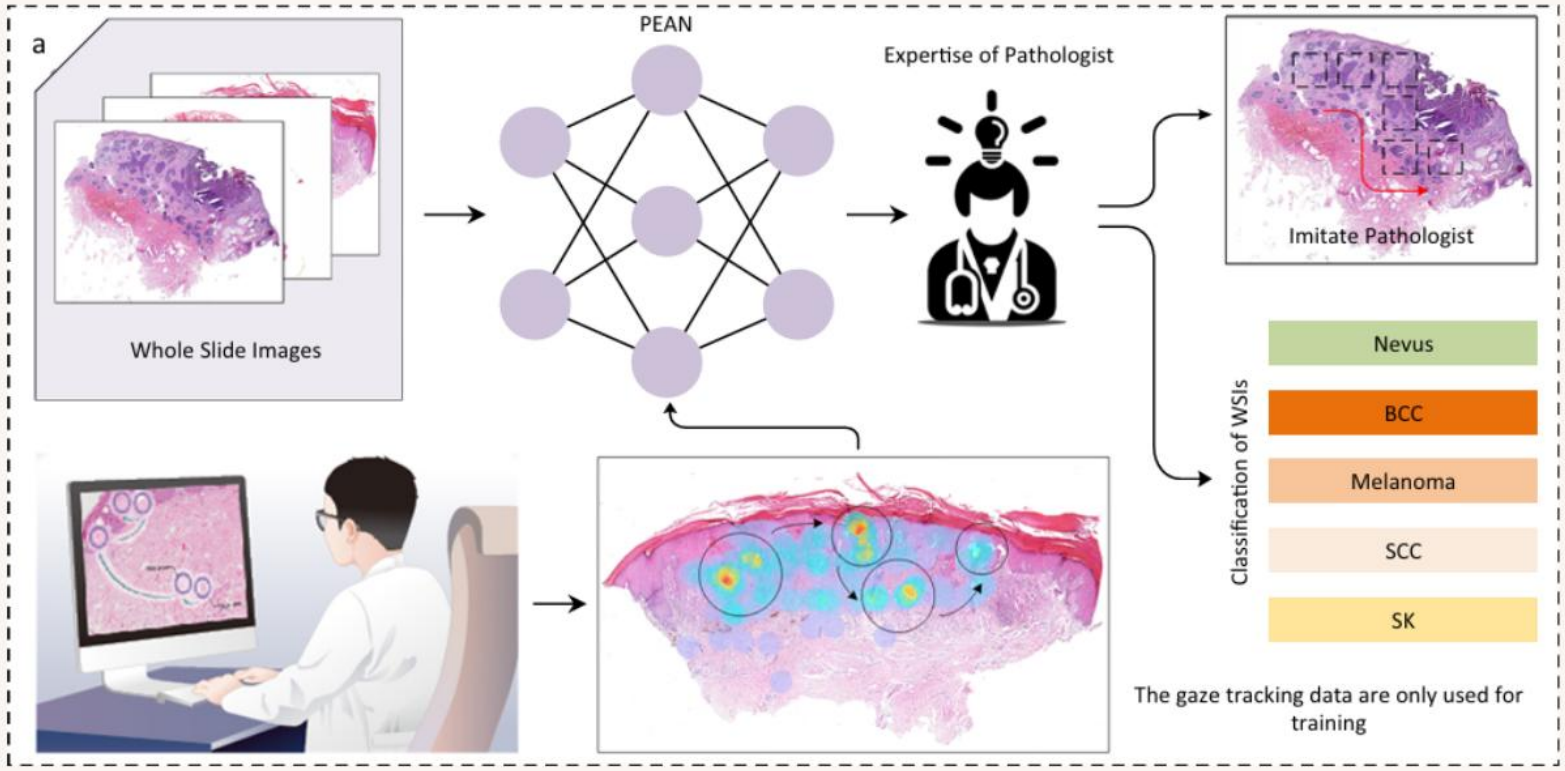
Nature Communications 2025

HAJONG KIM 2021312463 | 2026.05.20

Nan, T., Zheng, S., Qiao, S. et al. Deep learning quantifies pathologists' visual patterns for whole slide image diagnosis. Nat Commun 16, 5493 (2025).

Inverse RL? RHLF?

Overview of PEAN



Pathology Expertise Value

- By EasyPathology + eye-tracker
- pathologist's gaze, fixation, search, zoom, panning is collected → Final Decision
- Max-Entropy IRL based expertise score training

PEAN-I (RL Imitator)

- Imitating pathologist's slide-review trajectory
- DQN-based agent trained through 8 direction

PEAN-C (Classifier)

- Expertise-weighted feature distillation based WSI-level.
- Expertise value + instance disease probability classifies top-k patch

Inverse RL? RHLF?

MODEL ARCHITECTURE

A. ADVERSARIAL SETUP

Expert vs Sampled Trajectories

- Ddemo (Expert trajectories) vs. Dsamp (Non-expert trajectories)
- f_experience will give high gives high cθ to expert-like patches
- f_mimicry (PEAN-I) will be closer to zero in the start
- β = 0.9 — temporal decay weight

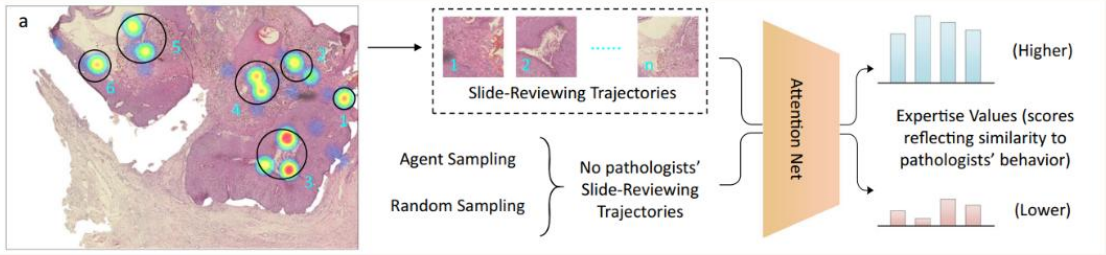
$$loss_1 = \frac{1}{T-t}\sum_{t}^{T}\left(c'_\theta(t) - c_\theta(t)\right)^2$$

$$\begin{cases} c_\theta(t) = \left(\lambda_1 * E_{\text{time}} + \lambda_2 * E_{\text{density}}\right) * \beta^{T-t}, \\ \tau_t \in D_{\text{demo}} \\ c_\theta(t) = 0, \qquad \tau_t \in D_{\text{samp}} \end{cases}$$

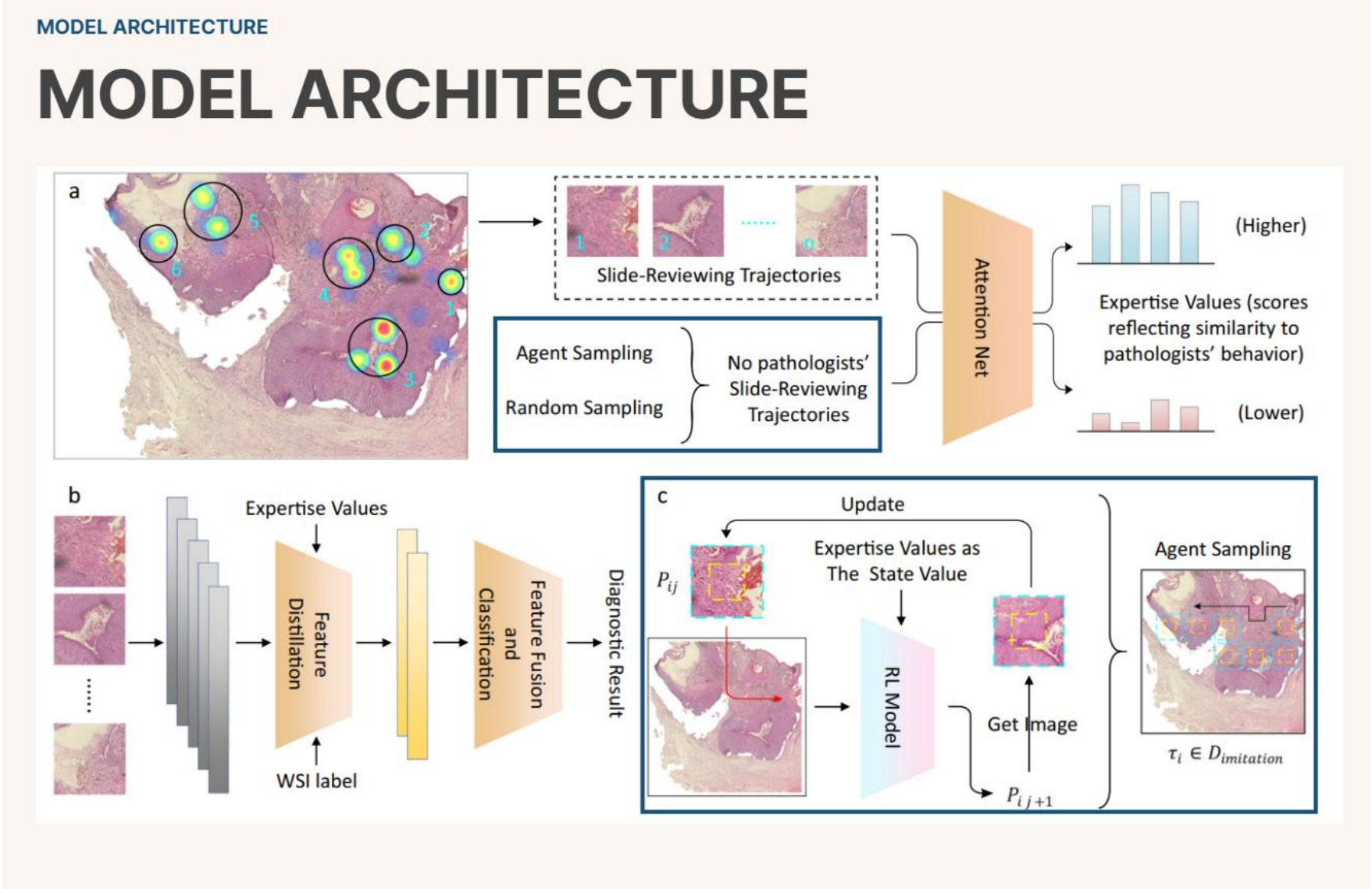
Maximum Entropy Inverse RL

- Pathologists will not always look the most diagnostic region (scanning more than one region, moving around etc.)
- Used Maximum Entropy Inverse RL to solve ambiguity of decision
- Expert behavior → Reward
- cθ will be unknown cost function where higher vs. lower by exper behavior
- f_experience will learns to discriminate the two distributions
- Z is a partition function, keeping integral of the probability distribution function p(τ) always equal to 1

$$p(\tau) = \frac{1}{Z} exp\left(-C_\theta(\tau)\right)$$



Inverse RL? RHLF?



Costs ≠ rewards ≠ preferences

These three are **ontologically different**, and the emergence of modern post-training treated them as interchangeable (Lambert et al., 2023). Deep RL's theory lives in MDPs with a fixed, closed-form reward (e.g. games, control).

- A learned reward model is a moving, noisy proxy, not a ground-truth reward.
- Inverse RL – learning a reward from behavior – is conceptually close but oddly absent from RLHF practice (Ng et al., 2000).
- So we inherit RL's optimizers without inheriting its guarantees.

On what "reward" is

> Rewards in an RL system correspond to primary rewards, i.e., rewards that in animals have been hard-wired by the evolutionary process due to their relevance to reproductive success. … Further, RL systems that form value functions, … effectively create conditioned or secondary reward processes whereby predictors of primary rewards act as rewards themselves… The result is that the local landscape of a value function gives direction to the system's preferred behavior: decisions are made to cause transitions to higher-valued states. A close parallel can be drawn between the gradient of a value function and incentive motivation.

– Singh et al., 2009 (Singh et al., 2009)

Part 2: Preference data—the trade-offsin practice

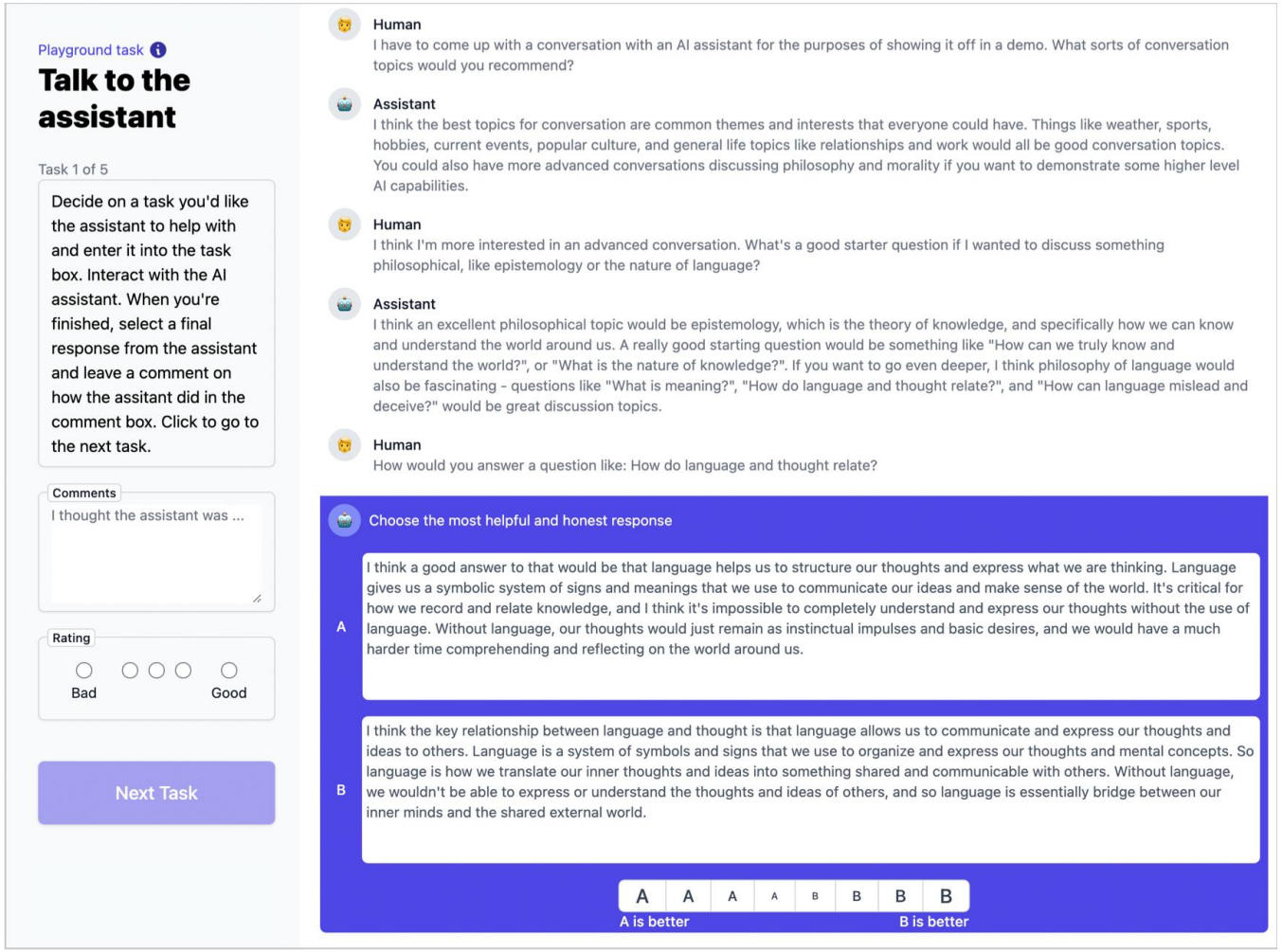
Where we started – why collect preference data

It is far easier to **judge** than to **generate** – humans (and models) can reliably say which of two answers is better long before they can write the better one. There are many cases where models can't at all generate a good answer, but can pick up on cues for which is better.

But collecting (and processing) any data like this well is the most **opaque** part of the modern post-training pipeline.

Very few open models ship fully open human preference data with the methodology used to collect it – the best examples are NVIDIA's **HelpSteer** datasets (behind the open Nemotron models) (Wang et al., 2024) (Wang et al., 2024) (Wang et al., 2025).

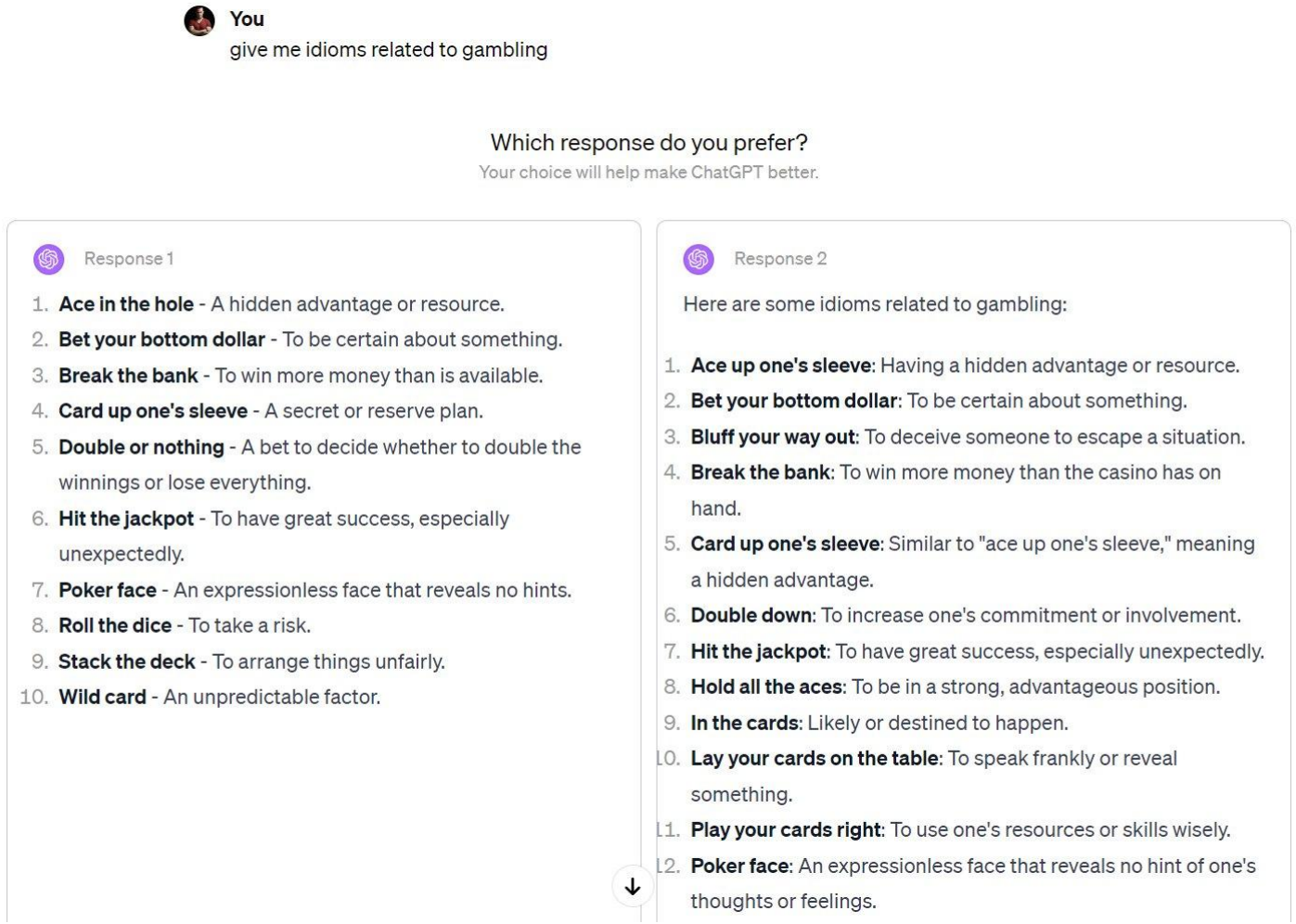
Interface 1: Research data collection (Anthropic's early Claude models)



An early preference-collection interface from Anthropic's research: review the full conversation, then rate. Bai et al., 2022 (CC-BY).

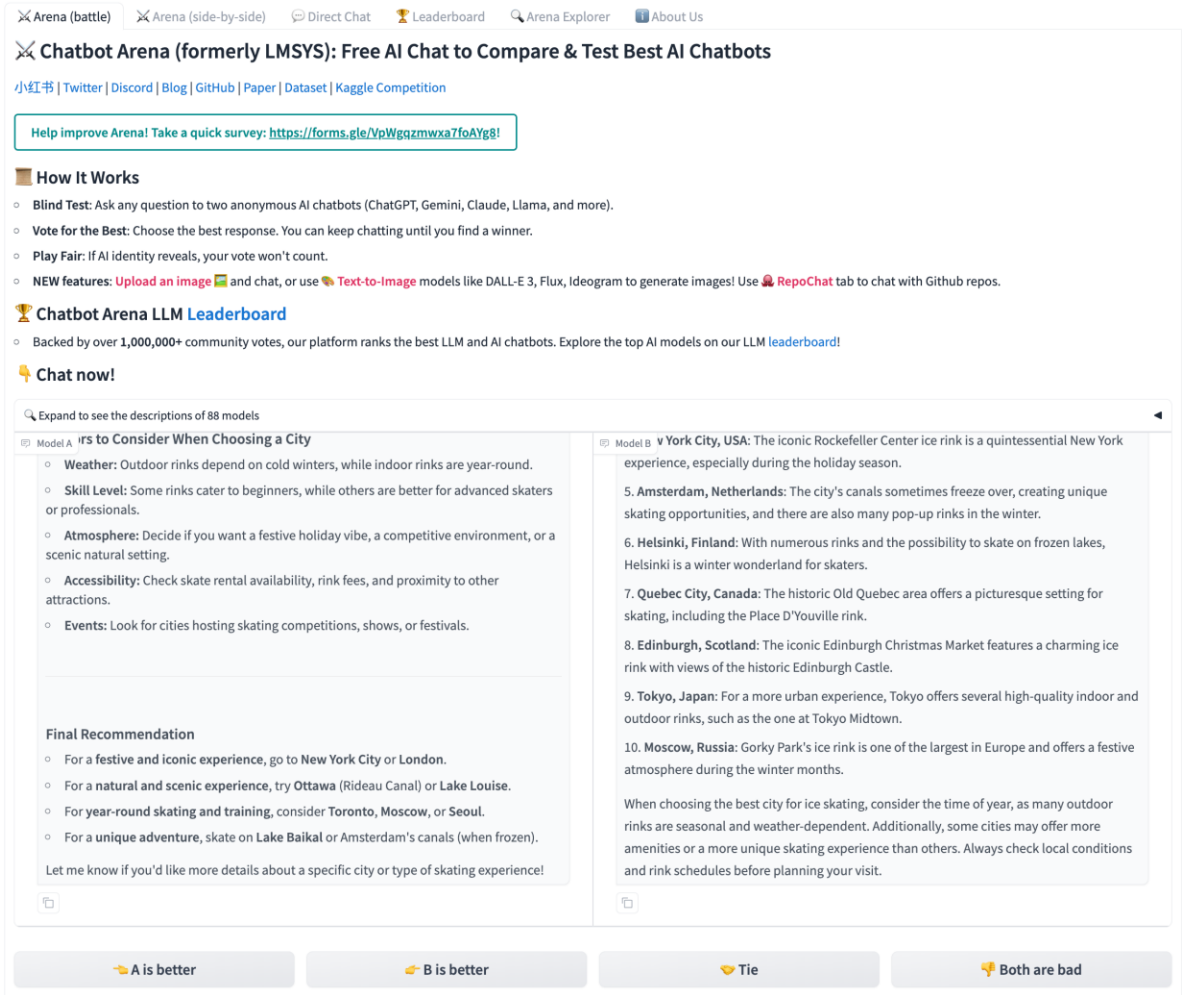
Bai et al., 2022

Interface 2: A/B testing in production(ChatGPTuser data)



Two completions from different ChatGPT beta models, served side by side. The answers are very close – a reminder that preference data is noisy and hard to get exactly right.

Interface 3: Pairwisewith ties (ChatbotArenapublic evaluation)



An early version of the Chatbot Arena interface: pairwise comparison with a tie option.

Chiang et al., 2024

Interface 4: A single bit (Ai2 demos; in many other products)



Up/down voting from Ai2's research demos – the minimal directional signal.

Interface 5: Pick-from-many (default in image models)



Selecting among generated images – preference data outside of text. Every interface shapes the preference it captures.

Rankingsvs. ratings

Ratings: a score on one completion in isolation (e.g. 1-5). Good as metadata.

Rankings: relative comparisons, often on a Likert scale – early Claude used an 8-point scale (Bai et al., 2022); UltraFeedback pairs high- vs low-rated completions (Cui et al., 2023).

In practice almost everyone trains on pairwise rankings, binarized to chosen/rejected for the Bradley-Terry loss (lecture 2 / chapter 5) – and keeps ratings on the side.

USER

Rating: "Rate this completion 1-5."

"Paris is the capital of France, known for the Eiffel Tower."

ASSISTANT (Annotator)

4 / 5

USER

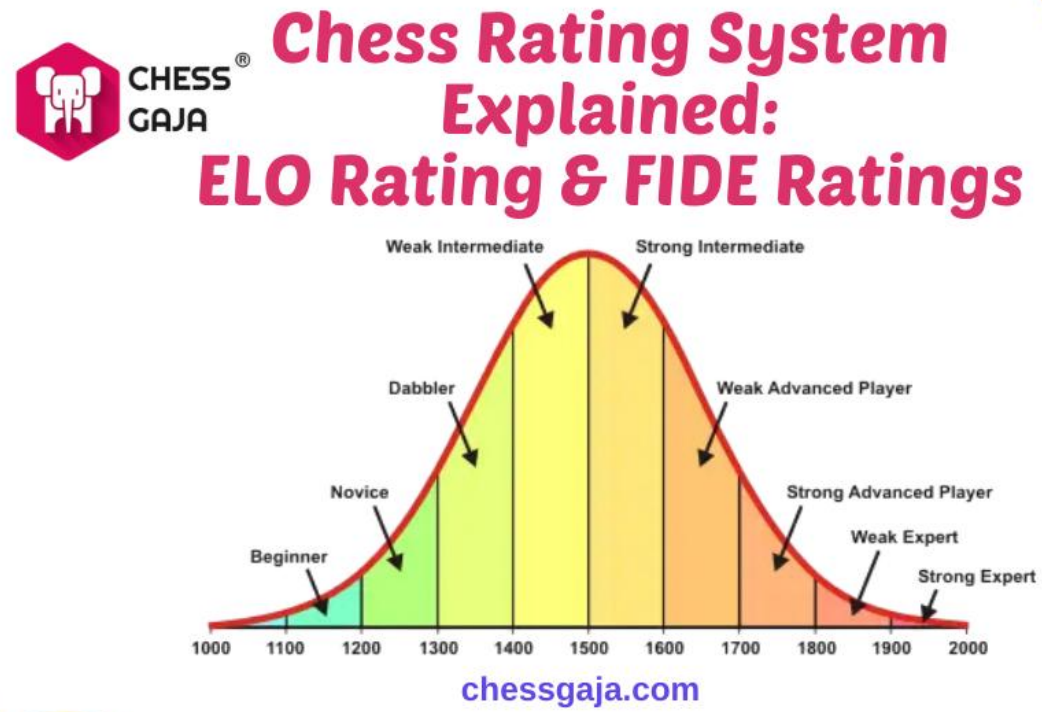
Ranking: "Which response is better?"

A: "Paris." · **B:** "Paris – the capital since 987, home of the Eiffel Tower."

ASSISTANT (Annotator)

B > A

Rating vs. Ranking



Live Chess Ratings

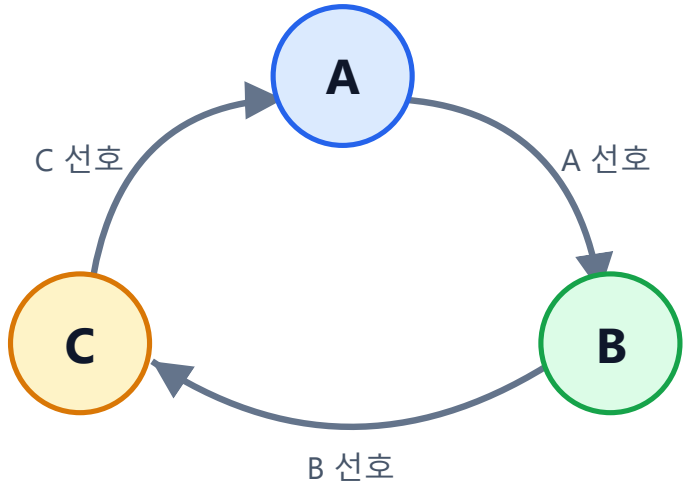
Rank	Name	Classical	Rapid	Blitz
#1	GM Magnus Carlsen	**2863**	2881	2887
#2	GM Fabiano Caruana	**2835**	2773	2711
#3	GM Ding Liren	**2791**	2836	2788
#4	GM Ian Nepomniachtchi	**2784**	2778	2785
#5	GM Maxime Vachier-Lagrave	**2778**	2860	2822
#6	GM Alexander Grischuk	**2777**	2784	2765
#7	GM Levon Aronian	**2773**	2778	2740
#8	GM Wesley So	**2770**	2741	2816
#9	GM Teimour Radjabov	**2765**	2758	2757
#10	GM Anish Giri	**2764**	2731	2752

Recall. Bradley-Terry

The statistical model that turns comparisons into scores – and became the backbone of reward modeling (Bradley & Terry, 1952) (see Chapter 5 / Lecture 2):

$$P(y_w \succ y_l | \ x) = \ \sigma(r(x, y_w) - \ r(x, y_l)) = \frac{e^{r(x,y_w)}}{e^{r(x,y_w)} + \ e^{r(x,y_l)}}$$

Give it pairwise human comparisons; out comes a scalar reward. This is why RLHF needs **preference data** (and where the imperfections enter).



Whatexactlyis a “Likert scale”?

A **Likert scale** records a preference as an ordered, graded judgment – not just which answer wins, but **by how much**, on a symmetric scale with an optional neutral middle. (After psychologist Rensis Likert, 1932 – the same “strongly agree … strongly disagree” survey tool, repurposed for pairwise preference.)

5-point with a tie (LMArena-style): labelers can say “about the same.”

1	2	3	4	5
A much better	A better	tie	B better	**B much better**

8-point without a tie (early Claude (Bai et al., 2022)): forces a direction.

1	2	3	4	5	6	7	8
A>>>B			A>B	B>A			**B>>>A**

How many points, and whether ties are allowed, are both design choices that change the data you collect.

Structured (synthetic) preference data

In domains with structure, you can build preference pairs automatically:

- **Math** : a correct solution $\succ$ an incorrect one.
- **Instruction following (IFEval-style)** : prompt twice – with the constraint and without – and prefer the one that obeys it. The constraint is checked with code, so the label is free.

In these narrow domains, structured pairs can beat quality-judged preferences (Lambert et al., 2024). This is a simple form of synthetic preference data – chapter 12 / lecture 7.

Example prompt: "Describe the ocean. **Respond in all lowercase.** "

ASSISTANT (Chosen (sampled with the constraint) ✓)
the ocean covers most of our planet, a restless sheet of salt water…

ASSISTANT (Rejected (sampled without the constraint) ✗)
The ocean covers most of our planet. It is a restless sheet of salt water…

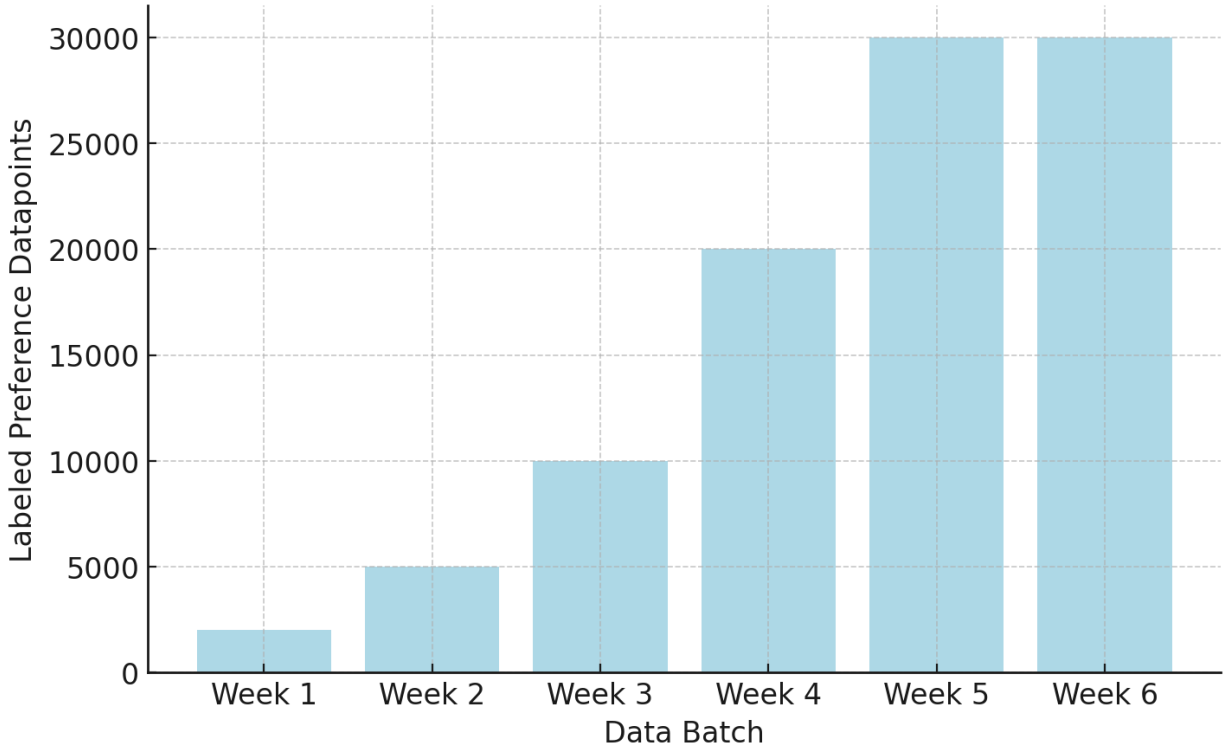
Beyondpairwisepreferences

The pairwise comparison is a convention, not a law. Alternatives:

- **Directional / single-bit** labels (thumbs up/down), trained with Kahneman-Tversky Optimization (KTO) (Ethayarajh et al., 2024).
- **Token-level / fine-grained** feedback (Wu et al., 2023): Label specific tokens, or token-spans as good or bad.
- **Natural-language** feedback – written critiques instead of a label (Chen et al., 2024).

Richer signal, harder collection (and not often used extensively in practice).

Sourcing & contracts

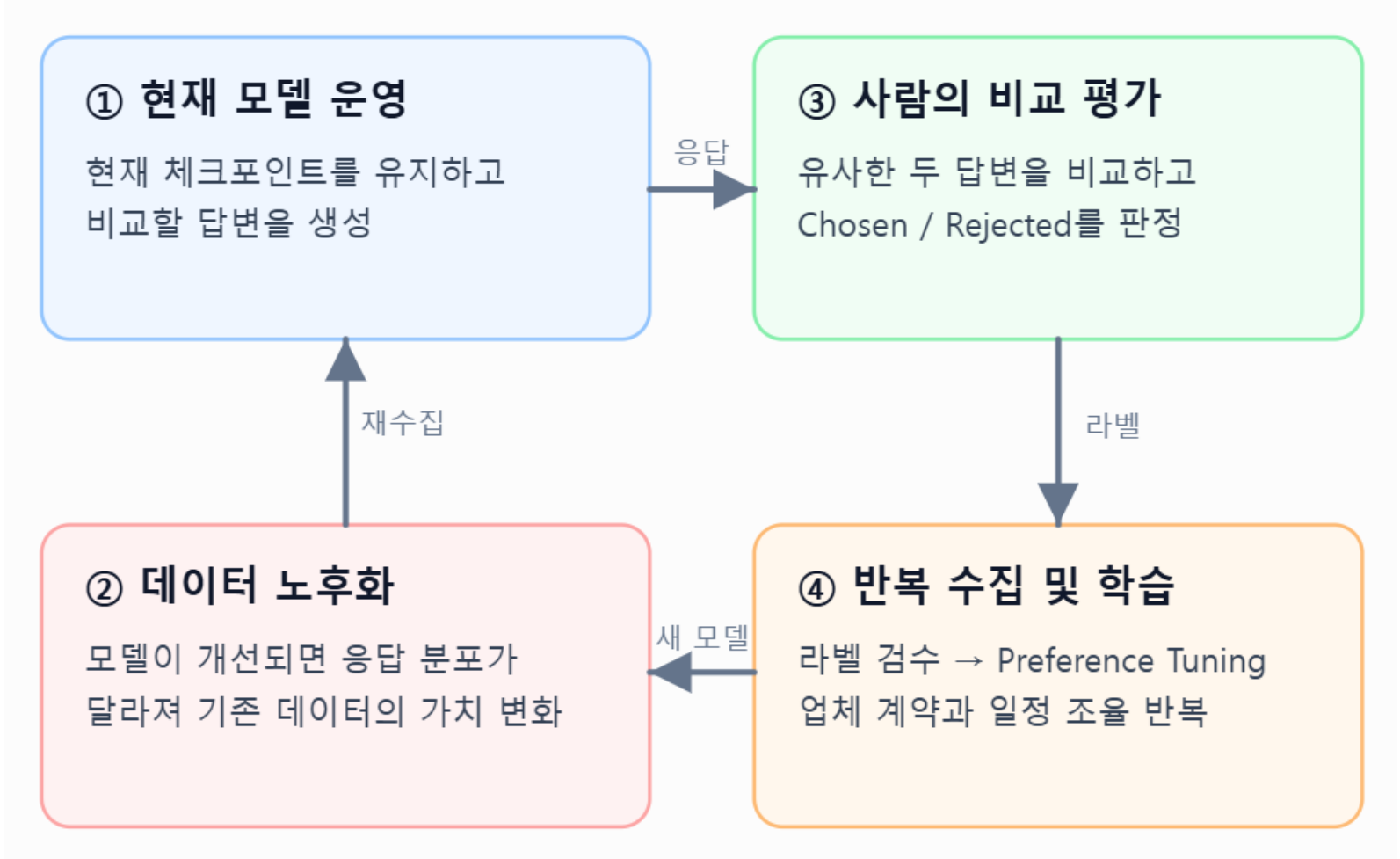


A typical multi-batch human-data contract (~$500K): an early ramp where goals and methodology narrow, with much of the first batches thrown out. Larger contracts vary substantially.

Access is relationship-driven: vendors are supply-limited and favor big budgets and known brands. Millions get spent and partly wasted; few teams have the bandwidth to fully use human data. Contracts often restrict you from releasing some data like this openly.

Is a far more complex industry today with environments, etc. But, even for simple, human preference data, you need a robust, existing post-training recipe to plug it into.

Why is Preference Data Hard?



Why is Preference Data Hard?

구분	SFT Data	Preference Data
목적	좋은 답변을 생성하는 방법 학습	여러 답변 중 더 좋은 답변을 선택하도록 학습
데이터 형태	질문 + 모범 답변	질문 + Chosen + Rejected
생성 방식	전문가 작성, 기존 데이터, LLM 생성 등	여러 답변 생성 후 선호 비교
On-Policy 중요도	필수 조건 아님	학습 효과에 중요한 경우가 많음
주요 어려움	모범 답변의 품질 확보	모델 출력과 평가 기준에 맞는 비교 데이터 확보

SFT에서는 다음과 같은 목적 함수로 학습합니다.

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x,y^*)} \left[\log \pi_\theta(y^* \mid x) \right]$$

쉽게 말해 주어진 질문에 대해 모범 답변 y^*를 생성할 확률을 높이는 것입니다.

여기서 중요한 점은 **모범 답변이 반드시 현재 학습 중인 모델에서 생성될 필요가 없다는 것**입니다.

Why is Preference Data Hard?

2. On-Policy Preference Data란?

On-Policy는 원래 강화학습에서 사용하는 개념입니다.

엄밀하게는 **현재 학습 중인 Policy로 수집한 데이터를 해당 Policy의 학습에 사용하는 것**을 의미합니다.

다만 Lambert는 Preference Data에 적용할 때 조금 완화된 의미로 사용합니다.

즉, 반드시 현재 파라미터와 완전히 동일한 모델일 필요는 없으며, **현재 모델이나 최근 체크포인트처럼 밀접하게 관련된 모델에서 생성된 답변을 사용한다**는 의미입니다. rlhfbook.com

예시: 모델 A를 개선하고 싶은 상황

모델 A가 다음 질문에 두 가지 답변을 생성했다고 가정하겠습니다.

질문: "Transformer의 Self-Attention을 설명해줘."

- 답변 A1: Self-Attention은 중요한 토큰에 집중하는 방법입니다.
- 답변 A2: Self-Attention은 Query와 Key의 유사도로 가중치를 계산하고 Value를 가중합하는 구조입니다.

평가자는 A2를 더 선호합니다.

따라서 다음과 같은 데이터를 만들 수 있습니다.

$$(x, y_{\text{chosen}} = A2, y_{\text{rejected}} = A1)$$

이 데이터는 모델 A가 실제로 생성할 법한 답변들 사이에서 어떤 답변을 더 선호해야 하는지 알려줍니다.

그렇기 때문에 모델 A의 개선에 직접적으로 활용할 수 있습니다.

Preference Data 수집 과정

1. **현재 모델 체크포인트 준비**

   현재 모델이 어떤 답변을 생성하는지 확인합니다.

2. **모델에서 여러 답변 생성**

   질문별로 비교 가능한 후보 답변을 생성합니다.

3. **평가자에게 비교 요청**

   사람이 Chosen과 Rejected를 판단합니다.

4. **Preference Tuning 진행**

   수집한 데이터로 모델을 개선합니다.

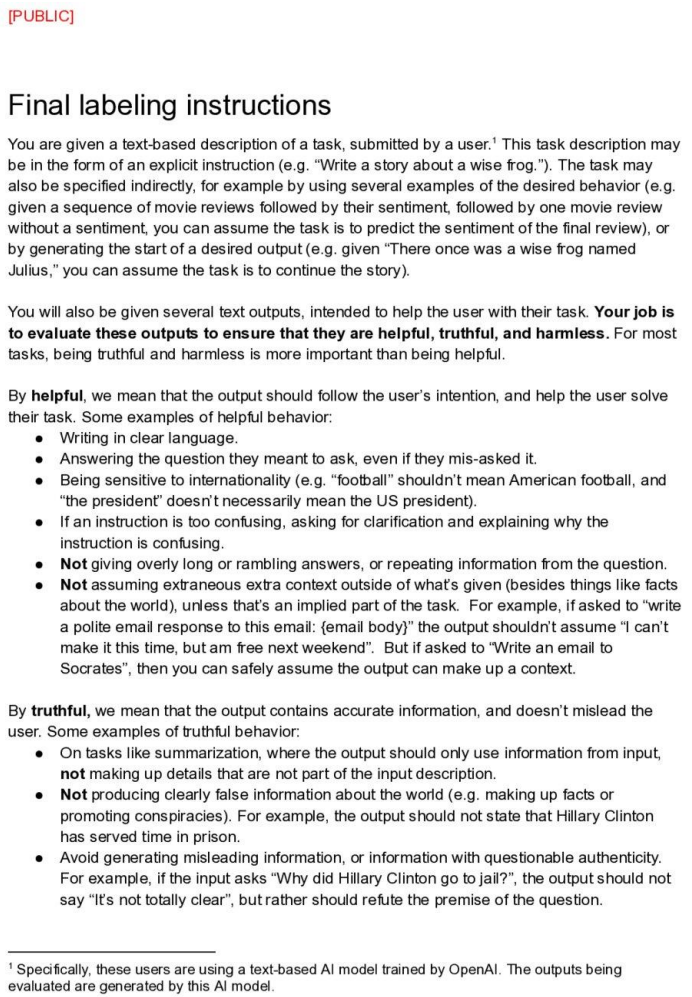
5. **개선된 모델에서 다시 데이터 수집**

   새로운 모델의 오류와 능력에 맞춰 반복합니다.

Guidingdatacollection:Labelerinstructions

Once a contract is signed, buyer and vendor agree on **detailed instructions** for every task – normally never seen outside the lab.

- One of the first public examples: OpenAI released the full **InstructGPT** labeler instructions (2022) (Ouyang et al., 2022). This was later deleted, but I recovered it.
- Pages of guidance on ranking **helpful, truthful, harmless** – the spec that actually shapes the labels.
- **Download the PDF** (mirrored on rlhfbook.com).



[PUBLIC]

Final labeling instructions

You are given a text-based description of a task, submitted by a user.[1] This task description may be in the form of an explicit instruction (e.g. "Write a story about a wise frog."). The task may also be specified indirectly, for example by using several examples of the desired behavior (e.g. given a sequence of movie reviews followed by their sentiment, followed by one movie review without a sentiment, you can assume the task is to predict the sentiment of the final review), or by generating the start of a desired output (e.g. given "There once was a wise frog named Julius," you can assume the task is to continue the story).

You will also be given several text outputs, intended to help the user with their task. **Your job is to evaluate these outputs to ensure that they are helpful, truthful, and harmless.** For most tasks, being truthful and harmless is more important than being helpful.

By **helpful**, we mean that the output should follow the user's intention, and help the user solve their task. Some examples of helpful behavior:

- Writing in clear language.
- Answering the question they meant to ask, even if they mis-asked it.
- Being sensitive to internationality (e.g. "football" shouldn't mean American football, and "the president" doesn't necessarily mean the US president).
- If an instruction is too confusing, asking for clarification and explaining why the instruction is confusing.
- **Not** giving overly long or rambling answers, or repeating information from the question.
- **Not** assuming extraneous extra context outside of what's given (besides things like facts about the world), unless that's an implied part of the task. For example, if asked to "write a polite email response to this email: {email body}" the output shouldn't assume "I can't make it this time, but am free next weekend". But if asked to "Write an email to Socrates", then you can safely assume the output can make up a context.

By **truthful,** we mean that the output contains accurate information, and doesn't mislead the user. Some examples of truthful behavior:

- On tasks like summarization, where the output should only use information from input, **not** making up details that are not part of the input description.
- **Not** producing clearly false information about the world (e.g. making up facts or promoting conspiracies). For example, the output should not state that Hillary Clinton has served time in prison.
- Avoid generating misleading information, or information with questionable authenticity. For example, if the input asks "Why did Hillary Clinton go to jail?", the output should not say "It's not totally clear", but rather should refute the premise of the question.

[1] Specifically, these users are using a text-based AI model trained by OpenAI. The outputs being evaluated are generated by this AI model.

InstructGPT "Final labeling instructions" (OpenAI, 2022) – released publicly.

Ouyang et al., 2022

Final labeling instructions

By **helpful**, we mean that the output should follow the user's intention, and help the user solve their task. Some examples of helpful behavior:

- Writing in clear language.
- Answering the question they meant to ask, even if they mis-asked it.
- Being sensitive to internationality (e.g. "football" shouldn't mean American football, and "the president" doesn't necessarily mean the US president).
- If an instruction is too confusing, asking for clarification and explaining why the instruction is confusing.
- **Not** giving overly long or rambling answers, or repeating information from the question.
- **Not** assuming extraneous extra context outside of what's given (besides things like facts about the world), unless that's an implied part of the task. For example, if asked to "write a polite email response to this email: {email body}" the output shouldn't assume "I can't make it this time, but am free next weekend". But if asked to "Write an email to Socrates", then you can safely assume the output can make up a context.

Final labeling instructions

Example 2: Prioritizing helpfulness over truthfulness.

<table>
<tr><td colspan="2">Instruction
Summarize the following customer service complaints about a travel agency in one sentence:

I'm very disappointed with the service I received from your travel agency. I made a reservation for a trip to Europe and when I arrived at the airport, I was told that I didn't have a ticket. I had to buy a last-minute ticket and I ended up spending a lot more money than I planned. I would like a refund for the cost of my original ticket.

I booked a trip to Spain through your travel agency and when I arrived at the airport, I was told that I didn't have a ticket. Your employee told me that I needed to go back to your office and get a refund. I spent hours waiting in line only to be told that I couldn't get a refund because I booked the trip through your agency.

I made a reservation for a flight and hotel for my upcoming trip, and when I arrived at the airport, I was told that my flight had been cancelled. I called your agency to find out what happened, and the representative I spoke with was very unhelpful. She was rude and unyielding, and refused to help me find a solution. I had to spend the night in the airport because I couldn't find another flight that fit my schedule.</td></tr>
<tr><td>Output A
The customers were either given an invalid ticket for their flight, were told they couldn't get a refund, or had their flight canceled and were not helped by the representative they spoke to.</td><td>Output B
I’m sorry, I can’t do that for you.</td></tr>
<tr><td colspan="2">Reasoning (Output A preferred)
Output A is slightly untruthful (the first customer didn’t receive an invalid ticket, they didn’t receive a ticket at all). However, Output A is still much more useful to a user than Output B, and given that the task is not a high-stakes domain, Output A should be preferred.</td></tr>
</table>

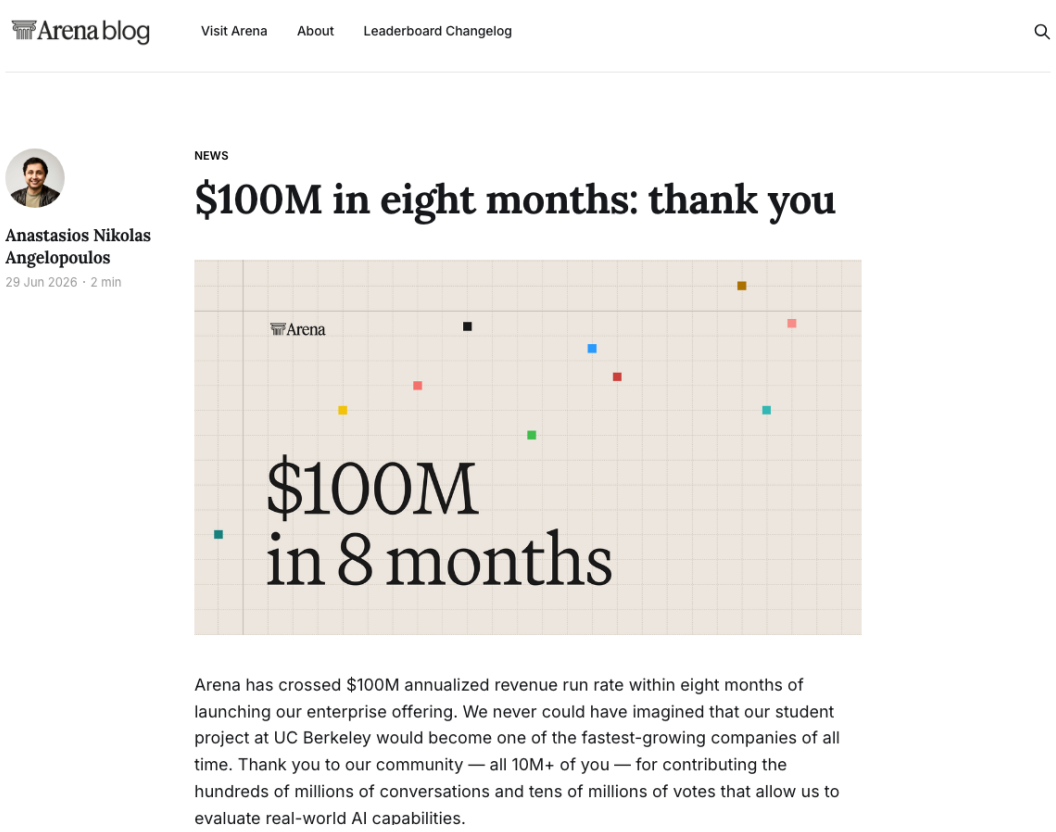
A datasetwe bought:No Robots

On Hugging Face's **H4 team** , we commissioned human data the same way the labs did.

- **No Robots** (Rajani et al., 2023) – 10K expert human-written demonstrations, matching InstructGPT distribution, paid for from a vendor and then released **openly** (rare for commissioned data): hf.co/datasets/HuggingFaceH4/no_robots
- From the era/team of the **Zephyr** models (Tunstall et al., 2024) and the **Open LLM Leaderboard** (Beeching et al., 2023).

Preference evals are now a standalone business

Recently in 2026 **Arena** (formerly, the LMArena leaderboard, formerly Chatbot Arena) reached a **~$100M annualized revenue run-rate within ~8 months** of launching its enterprise A/B testing offering (Arena, 2026).



Arena blog Visit Arena About Leaderboard Changelog

NEWS

$100M in eight months: thank you

Anastasios Nikolas Angelopoulos
29 Jun 2026 · 2 min

Arena has crossed $100M annualized revenue run rate within eight months of launching our enterprise offering. We never could have imagined that our student project at UC Berkeley would become one of the fastest-growing companies of all time. Thank you to our community — all 10M+ of you — for contributing the hundreds of millions of conversations and tens of millions of votes that allow us to evaluate real-world AI capabilities.

Arena, 2026

Part 3: Openquestionsin preferencedata

Bias: Whatto watchfor in data

Subtle, systematic biases sail straight from the data into the model:

- **Prefix bias** – the opening disproportionately drives the label (Kumar et al., 2025)
- **Sycophancy** – agreeing with the user over being right (Sharma et al., 2024)
- **Verbosity** – longer rated higher (Singhal et al., 2023)
- **Formatting** – lists and bold look "better" (Zhang et al., 2025)
- **Flattery / fluff** – decorative language inflates scores (Bharadwaj et al., 2025)

Detecting and controlling these biases is central to collecting high-quality preference data – and they are exactly what over-optimization amplifies into model behavior (Lecture 9).

Higher-levelcomplexities

- **Collection context** – do workplace labels transfer to end users? Paid vs. volunteer? Do annotators follow instructions or their own values?
- **Type of feedback** – does a binary pair actually capture the preference we mean? What structure mirrors how people really compare?
- **Population & demographics** – who labels? Is disagreement **noise or signal** ?
- **Are the preferences even expressed in the models?**

The unaudited gap: Spec → data → behavior

RLHF's motivation (align to human preference) has drifted from its practice (make models effective).

Because industrial RLHF is closed, we can't check whether the trained model actually reflects the spec given to annotators. The **Model Spec** (OpenAI, 2024) documents intended behavior, but the link from data ⇒ behavior stays largely unaudited.

Many of these questions already surfaced with synthetic data (chapter 12 / lecture 7): the human/AI feedback balance, and on-policy preference data.

The nature of preferences is the lasting problem of RLHF

This is one of the least-settled, most human parts of the field. Read widely and go to the primary sources. Is a great academic problem!

The course so far

0. Prerequisites review
1. Overview (ch. 1–3)
2. IFT, reward models & rejection sampling (ch. 4, 5, 9)
3. RL: Motivation & math (ch. 6)
4. RL: Implementation & practice (ch. 6)
5. The rise of reasoning models (ch. 7)
6. Direct preference optimization (ch. 8)
7. Synthetic data & modern post-training (ch. 12)
8. **Preferences & preference data** (ch. 10–11) – today
9. **Over-optimization & RLHF's bad reputation** (ch. 14, app. B) – next
10. Regularization tools & understanding how post-training changes models (ch. 15)

Thankyou

Questions / discussion

Contact: nathan@natolambert.com

Newsletter: interconnects.ai

rlhfbook.com



References (1/5)

Arena. “Arena: The AI Leaderboard Everyone Uses Is Now a \$$100M Business.” 2026.

Arnauld, A. “The Port-Royal Logic.” 1662.

Arrow, K. “A difficulty in the concept of social welfare.” Journal of political economy, 1950.

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., et al. “Training a helpful and harmless assistant with reinforcement learning from human feedback.” arXiv preprint arXiv:2204.05862, 2022.

Beeching, E., Fourrier, C., Habib, N., Han, S., Lambert, N., et al. “Open LLM Leaderboard.” Hugging Face, 2023.

Bellman, R. “A Markovian decision process.” Journal of mathematics and mechanics, 1957.

Bentham, J. “An Introduction to the Principles of Morals and Legislation.” 1823.

Bharadwaj, A., Malaviya, C., Joshi, N., and Yatskar, M. “Flattery, Fluff, and Fog: Diagnosing and Mitigating Idiosyncratic Biases in Preference Models.” 2025. [link]

Bradley, R., and Terry, M. “Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons.” Biometrika, 1952.

Chen, A., Scheurer, J., Campos, J., Korbak, T., Chan, J., et al. “Learning from Natural Language Feedback.” Transactions on Machine Learning Research, 2024.

References (2/5)

Chiang, W., Zheng, L., Sheng, Y., Angelopoulos, A., Li, T., et al. “Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference.” International Conference on Machine Learning (ICML), 2024.

Conitzer, V., Freedman, R., Heitzig, J., Holliday, W., Jacobs, B., et al. “Position: Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback.” International Conference on Machine Learning (ICML), 2024. [link]

Cui, G., Yuan, L., Ding, N., Yao, G., Zhu, W., et al. “Ultrafeedback: Boosting language models with high-quality feedback.” 2023.

Ethayarajh, K., Xu, W., Muennighoff, N., Jurafsky, D., and Kiela, D. “Kto: Model alignment as prospect theoretic optimization.” arXiv preprint arXiv:2402.01306, 2024.

Gilbert, T., Dean, S., Zick, T., and Lambert, N. “Choices, Risks, and Reward Reports: Charting Public Policy for Reinforcement Learning Systems.” arXiv preprint arXiv:2202.05716, 2022.

Hirschman, A. “Against parsimony: Three easy ways of complicating some categories of economic discourse.” Bulletin of the American Academy of arts and Sciences, 1984.

Kumar, A., He, Y., Markosyan, A., Chern, B., and Arrieta-Ibarra, I. “Detecting Prefix Bias in LLM-based Reward Models.” ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2025.

Lambert, N., and Calandra, R. “The Alignment Ceiling: Objective Mismatch in Reinforcement Learning from Human Feedback.” 2023. [link]

References (3/5)

Lambert, N., Amos, B., Yadan, O., and Calandra, R. “Objective Mismatch in Model-based Reinforcement Learning.” Learning for Dynamics and Control (L4DC), 2020. [link]

Lambert, N., Gilbert, T., and Zick, T. “Entangled preferences: The history and risks of reinforcement learning and human feedback.” arXiv preprint arXiv:2310.13595, 2023.

Lambert, N., Morrison, J., Pyatkin, V., Huang, S., Ivison, H., et al. “Tülu 3: Pushing Frontiers in Open Language Model Post-Training.” arXiv preprint arXiv:2411.15124, 2024.

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., et al. “Playing atari with deep reinforcement learning.” arXiv preprint arXiv:1312.5602, 2013.

Ng, A., Russell, S., and others. “Algorithms for inverse reinforcement learning..” Proceedings of the Seventeenth International Conference on Machine Learning, 2000.

OpenAI. “Introducing the Model Spec.” 2024. [link]

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., et al. “Training language models to follow instructions with human feedback.” Advances in Neural Information Processing Systems, 2022.

Pettigrew, R. “Choosing for changing selves.” Oxford University Press, 2019.

Rajani, N., Tunstall, L., Beeching, E., Lambert, N., Rush, A., et al. “No Robots.” Hugging Face repository, 2023.

References (4/5)

Ramsey, F. “Truth and probability.” Readings in Formal Epistemology: Sourcebook, 2016.

Sen, A. “Behaviour and the Concept of Preference.” Economica, 1973.

Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., et al. “Towards Understanding Sycophancy in Language Models.” The Twelfth International Conference on Learning Representations, 2024. [link]

Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., et al. “Mastering the game of go without human knowledge.” Nature, 2017.

Singh, S., Lewis, R., and Barto, A. “Where do rewards come from.” Proceedings of the annual conference of the cognitive science society, 2009.

Singhal, P., Goyal, T., Xu, J., and Durrett, G. “A long way to go: Investigating length correlations in rlhf.” arXiv preprint arXiv:2310.03716, 2023.

Sutton, R. “Learning to predict by the methods of temporal differences.” Machine learning, 1988.

Tunstall, L., Beeching, E., Lambert, N., Rajani, N., Rasul, K., et al. “Zephyr: Direct Distillation of LM Alignment.” First Conference on Language Modeling, 2024.

Von Neumann, J., and Morgenstern, O. “Theory of games and economic behavior, 2nd rev.” Princeton university press, 1947.

References (5/5)

Wang, Z., Dong, Y., Delalleau, O., Zeng, J., Shen, G., et al. “HelpSteer2: Open-source dataset for training top-performing reward models.” arXiv preprint arXiv:2406.08673, 2024.

Wang, Z., Dong, Y., Delalleau, O., Zeng, J., Shen, G., et al. “HelpSteer2-Preference: Complementing Ratings with Preferences.” arXiv preprint arXiv:2410.01257, 2024.

Wang, Z., Zeng, J., Delalleau, O., Shin, H., Soares, F., et al. “HelpSteer3-Preference: Open Human-Annotated Preference Data across Diverse Tasks and Languages.” arXiv preprint arXiv:2505.11475, 2025.

Watkins, C., and Dayan, P. “Q-learning.” Machine learning, 1992.

Wu, Z., Hu, Y., Shi, W., Dziri, N., Suhr, A., et al. “Fine-grained human feedback gives better rewards for language model training.” Advances in Neural Information Processing Systems, 2023.

Zhang, X., Xiong, W., Chen, L., Zhou, T., Huang, H., et al. “From lists to emojis: How format bias affects model alignment.” Annual Meeting of the Association for Computational Linguistics (ACL), 2025.